



# ARABICA

Coffee Data Set – V.1.0



# Contents

|                                                       |    |
|-------------------------------------------------------|----|
| Data Set Variables .....                              | 2  |
| Arabica Major Growing Regions .....                   | 3  |
| South America .....                                   | 5  |
| Brazil (1 <sup>st</sup> ) .....                       | 5  |
| Colombia (2 <sup>nd</sup> ).....                      | 5  |
| Peru (6 <sup>th</sup> ) .....                         | 6  |
| Central America / Caribbean .....                     | 7  |
| Honduras (4 <sup>th</sup> ) .....                     | 7  |
| Guatemala (5 <sup>th</sup> ).....                     | 8  |
| Nicaragua (8 <sup>th</sup> ) .....                    | 8  |
| Mexico (7 <sup>th</sup> ).....                        | 9  |
| Africa .....                                          | 9  |
| Ethiopia (3 <sup>rd</sup> ) .....                     | 9  |
| Asia .....                                            | 10 |
| China (9 <sup>th</sup> ) .....                        | 10 |
| India (10 <sup>th</sup> ).....                        | 10 |
| Summary .....                                         | 11 |
| Coordinates (Nasapower).....                          | 14 |
| Climate & Elevation Modelling.....                    | 14 |
| Visual Confirmation .....                             | 17 |
| Study Areas.....                                      | 20 |
| Minas Gerais - Brazil [32,48%].....                   | 21 |
| Central Western Colombia [9,89%].....                 | 23 |
| São Paulo – Brazil [6,50%] .....                      | 25 |
| Oromia – Ethiopia [5,90%] .....                       | 26 |
| Espírito Santo – Brazil [4,64%] .....                 | 28 |
| Former SNNPR – Ethiopia [2,48%] .....                 | 30 |
| Intibucá – Honduras [1,81%] .....                     | 30 |
| Olancho – Honduras [1,81%].....                       | 31 |
| El Paraíso Department – Honduras [1,81%].....         | 31 |
| Yunnan – China [1,80%] .....                          | 31 |
| Southern – Colombia [1,71%].....                      | 32 |
| San Marcos – Guatemala [1,39%].....                   | 32 |
| Huehuetenango – Guatemala [1,29%].....                | 32 |
| Jalapa – Guatemala [1,11%].....                       | 33 |
| Chialapas – Mexico [1,29%] .....                      | 33 |
| Cajamarca – Peru [0,98%] .....                        | 33 |
| Exchange Rate Information .....                       | 34 |
| Futures as Proxies for Arabica Spot Prices.....       | 34 |
| Conclusions and Streamlining Commodity Research ..... | 35 |

## Data Set Variables

| Variables        | Explanation                                                                                                                                                                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LON              | Longitude                                                                                                                                                                                                                                                                            |
| LAT              | Latitude                                                                                                                                                                                                                                                                             |
| YEAR             | -                                                                                                                                                                                                                                                                                    |
| DD               | Day                                                                                                                                                                                                                                                                                  |
| MM               | Month                                                                                                                                                                                                                                                                                |
| DOY              | Date of year, i.e, the sequential number of the day within a given year,                                                                                                                                                                                                             |
| Date             | Full Date                                                                                                                                                                                                                                                                            |
| Temp_Max         | The maximum hourly air (dry bulb) temperature at 2 meters above the surface of the earth in the period of interest (NASA POWER, T2M_MAX). <sup>1</sup>                                                                                                                               |
| Temp_Min         | The minimum hourly air (dry bulb) temperature at 2 meters above the surface of the earth in the period of interest (NASA POWER, T2M_MIN). <sup>1</sup>                                                                                                                               |
| Humidity         | The ratio of actual partial pressure of water vapor to the partial pressure at saturation, expressed in percent (NASA POWER, RH2M). <sup>1</sup>                                                                                                                                     |
| Solar_Radiation  | The total solar irradiance incident (direct plus diffuse) on a horizontal plane at the surface of the earth under all sky conditions. An alternative term for the total solar irradiance is the "Global Horizontal Irradiance" or GHI (NASA POWER, ALLSKY_SFC_SW_DWN ). <sup>1</sup> |
| Precipitation_mm | The bias corrected average of total precipitation at the surface of the earth in water mass (includes water content in snow). (NASA POWER, PRECTOTCORR). <sup>1</sup>                                                                                                                |
| Location         | Location of the area of study Country_Location (E.g. Colombia_Timbio)                                                                                                                                                                                                                |
| Close_USD_60kg   | Coffee Future Contract converted from cents per pounds to USD per 60kg, to reflect spot price.                                                                                                                                                                                       |
| Is Weekend       | Boolean as to whether the day is a weekend                                                                                                                                                                                                                                           |
| Is Interpolated  | Boolean as to whether the value for financial data has been interpolated due to a weekend or unexpected market closing.                                                                                                                                                              |
| Brl              | USD Spot Exchange Rate to Brazilian Real                                                                                                                                                                                                                                             |
| Cny              | USD Spot Exchange Rate to Chinese Yuan Renminbi.                                                                                                                                                                                                                                     |
| mxn              | USD Spot Exchange Rate to Mexican Pesos                                                                                                                                                                                                                                              |

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<sup>1</sup> Callejo, A. NASA POWER Cheatsheet. GitHub. Available at: <https://gist.github.com/abelcallejo/d68e70f43ffa1c8c9f6b5e93010704b8> [Accessed 3 Jul. 2025].

## Arabica Major Growing Regions

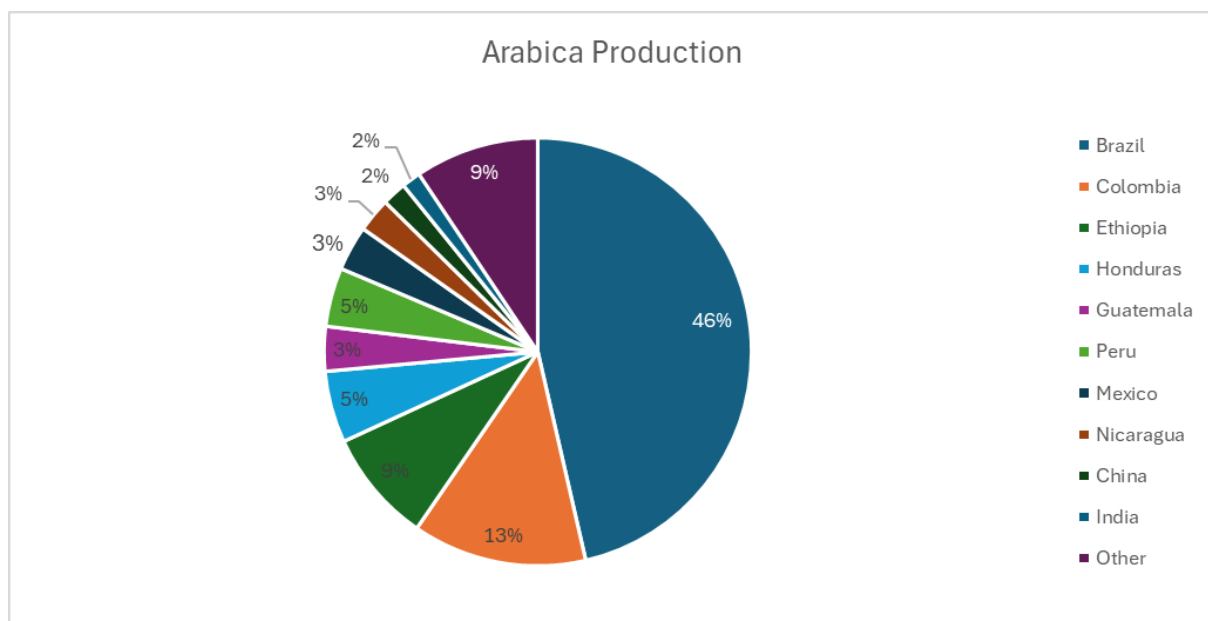
The original plan was to gather weather data for all Arabica-growing regions globally. However, this quickly proved to be a substantial task. Arabica is cultivated across a wide range of microclimates, even within the same country, requiring localized data. With 34 producing countries, many divided into four or more key regions, this would have meant collecting weather variables such as minimum and maximum temperature, humidity, solar radiation, and precipitation for nearly 100 regions. This amounts to over 500 weather variables, not including additional macroeconomic indicators like exchange rates. To maintain feasibility while preserving analytical power, the scope of the dataset was first narrowed to the top ten Arabica-producing regions, which together account for approximately 91% of global production, with multiple data points collected in major growing areas. This was revised to those areas that account for 79% of global production.

### Arabica Growing Regions by Thousand 60-Kilogram Bags<sup>2</sup>

| Country            | 2020/21 | 2021/22 | 2022/23 | 2023/24 | Jun 2024/25 | Dec 2024/25 |
|--------------------|---------|---------|---------|---------|-------------|-------------|
| Brazil             | 49700   | 36400   | 39800   | 44900   | 48200       | 45400       |
| Colombia           | 13400   | 11800   | 10700   | 12760   | 12400       | 12900       |
| Ethiopia           | 7600    | 8150    | 7300    | 8600    | 8360        | 8360        |
| Honduras           | 6500    | 4800    | 5700    | 5000    | 5300        | 5300        |
| Guatemala          | 3810    | 3410    | 3150    | 3300    | 3000        | 3300        |
| Peru               | 3369    | 4200    | 3475    | 4000    | 4250        | 4350        |
| Mexico             | 3000    | 3200    | 3000    | 3325    | 3350        | 3325        |
| Nicaragua          | 2550    | 2718    | 2430    | 2200    | 2525        | 2525        |
| China              | 1900    | 1800    | 1800    | 1800    | 1900        | 1800        |
| India              | 1650    | 1583    | 1667    | 1480    | 1400        | 1400        |
| Costa Rica         | 1472    | 1215    | 1100    | 1100    | 1100        | 1100        |
| Indonesia          | 1300    | 1280    | 1200    | 1350    | 1400        | 1400        |
| Vietnam            | 950     | 1100    | 1000    | 1000    | 1150        | 1100        |
| Uganda             | 730     | 990     | 990     | 1000    | 1000        | 1000        |
| Kenya              | 693     | 860     | 750     | 775     | 750         | 750         |
| Papua New Guinea   | 650     | 725     | 825     | 930     | 825         | 915         |
| Tanzania           | 600     | 650     | 650     | 750     | 775         | 775         |
| El Salvador        | 540     | 600     | 550     | 500     | 450         | 500         |
| Venezuela          | 480     | 500     | 500     | 500     | 500         | 500         |
| Rwanda             | 300     | 270     | 260     | 250     | 275         | 275         |
| Burundi            | 230     | 165     | 175     | 140     | 215         | 150         |
| Dominican Republic | 120     | 125     | 125     | 125     | 125         | 125         |
| Cuba               | 100     | 100     | 100     | 100     | 100         | 100         |
| Ecuador            | 88      | 90      | 113     | 120     | 105         | 105         |
| Bolivia            | 80      | 85      | 90      | 70      | 100         | 85          |
| Congo (Kinshasa)   | 75      | 70      | 75      | 75      | 75          | 75          |
| Panama             | 75      | 75      | 85      | 75      | 75          | 75          |
| Cameroon           | 50      | 25      | 50      | 25      | 25          | 25          |
| United States      | 33      | 38      | 40      | 50      | 50          | 50          |
| Madagascar         | 25      | 25      | 25      | 25      | 25          | 25          |
| Philippines        | 25      | 25      | 25      | 25      | 25          | 25          |
| Jamaica            | 15      | 15      | 18      | 19      | 15          | 20          |
| Malawi             | 10      | 11      | 11      | 10      | 10          | 10          |

<sup>2</sup> United States Department of Agriculture Foreign Agricultural Service, Coffee: World Markets and Trade (Washington, DC: USDA FAS, December 2024).

## Country Production as share of total production<sup>1</sup>



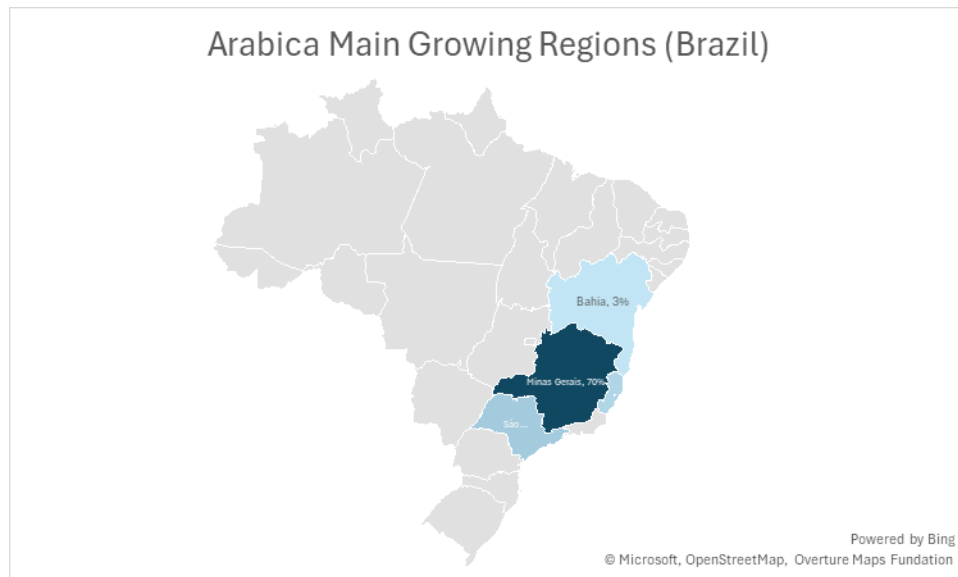
It is important to note that not all Arabica is created equal. Regional differences in cultivar, processing methods, and quality mean that certain markets may exhibit characteristics of monopsony or price segmentation for specific types of Arabica. Moreover, not all coffee is harvested during the same period, even within the same country.

| Harvesting Season | Country                                                                                                                                                                                                                                                                                                                       |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| April-March       | Angola, Bolivia, Burundi, Ecuador, Indonesia<br>Madagascar, Papua New Guinea, Peru, Rwanda                                                                                                                                                                                                                                    |
| July-June         | Brazil, Cuba, Dominican Republic, Haiti,<br>Philippines, Tanzania                                                                                                                                                                                                                                                             |
| October-September | Cameroon Central African Republic China<br>Colombia Congo (Kinshasa) Costa Rica Cote<br>d'Ivoire El Salvador Ethiopia Ghana Guatemala<br>Guinea Honduras India Jamaica Kenya Laos<br>Liberia Malawi Malaysia Mexico Nicaragua<br>Nigeria Panama Sierra Leone Thailand Togo<br>Uganda United States Venezuela Vietnam<br>Yemen |

Source: USDA<sup>1</sup>

## South America

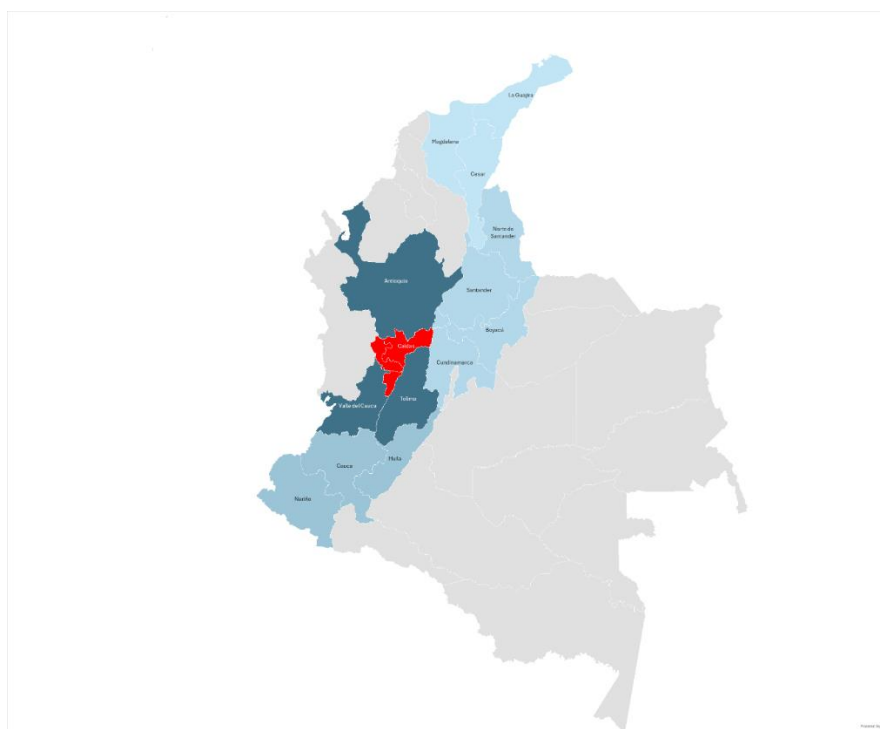
### Brazil (1<sup>st</sup>)



Brazil account for roughly half of Arabica global production, which is concentrated in four major regions: Minas Gerais [70%]; São Paulo [14%]; Espírito Santo [10%]; and Bahia [3%].<sup>3</sup>

### Colombia (2<sup>nd</sup>)

#### Arabica Main Growing Regions (Colombia)



<sup>3</sup> Marcela Formiga, Coffee Annual: Brazil (Brasilia: USDA Foreign Agricultural Service, 3 June 2024), Report No. BR2024-0012, approved by Joseph Degreenia.

Most of the Colombian coffee is grown in the so-called Coffee-Growing Axis (Showned in Red), consisting of the departments Caldas, Quindío and Risaralda.

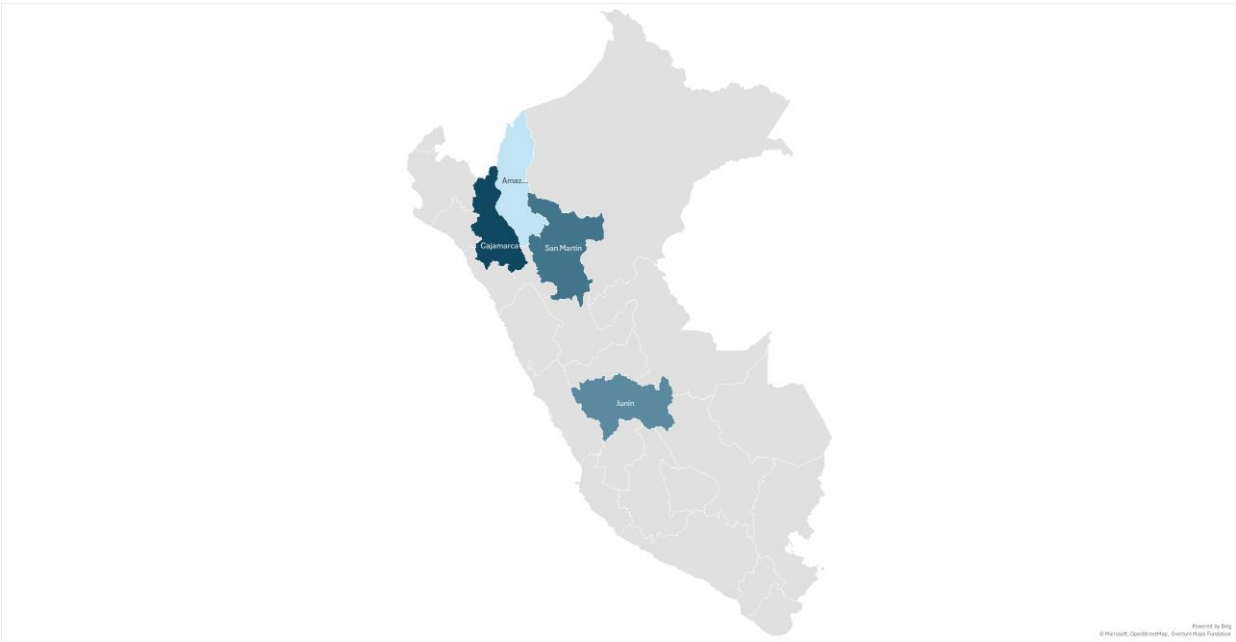
| REGION                   | AREA PLANTED | COFFEE PRODUCTION | REGION                                                                 |
|--------------------------|--------------|-------------------|------------------------------------------------------------------------|
| CENTRAL WESTERN SOUTHERN | 61%          | 75%               | Antioquia, Caldas, Quindío, Risaralda, Tolima and Valle                |
| EASTERN                  | 18%          | 13%               | Southern Region, comprising the departments of Cauca, Nariño and Huila |
| NORTHERN                 | 16%          | 9%                | Boyacá, Cundinamarca, Norte de Santander and Santander;                |
|                          | 5%           | 3%                | Cesar, La Guajira and Magdalena.                                       |

Note, the table above is sourced from 2002<sup>4</sup>. However, more recent data confirms that the Central-Western region of Colombia remains the country’s dominant Arabica-producing area, largely due to the enduring strength of the Eje Cafetero (Coffee-Growing Axis). Notably, Caldas alone accounted for 10.33% of Colombia’s national green coffee bean production<sup>5</sup>.

Peru (6<sup>th</sup>)

Coffee in Peru, which consist solely of arabica, is produced throughout the eastern slope of the Andes, with the main regions being Cajamarca [22%], San Martin [20%], Junin [19%] percent and Amazonas [15%].<sup>6</sup>

Arabica Main Growing Regions (Peru)



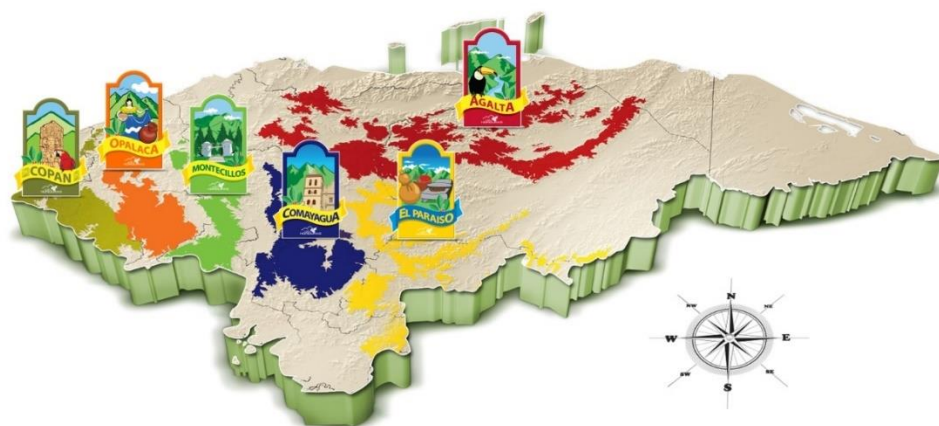
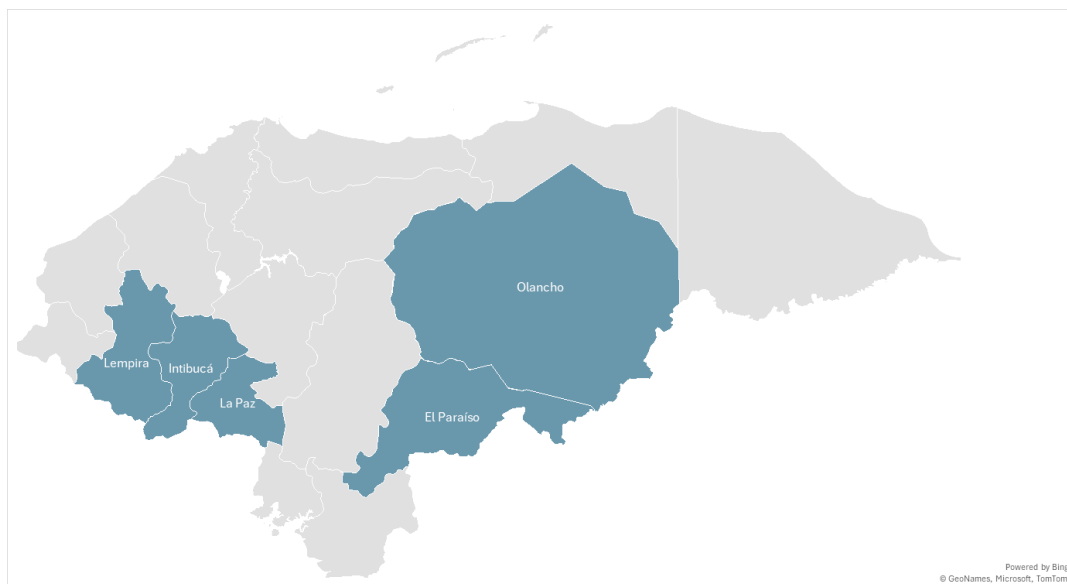
<sup>4</sup> Daniele Giovannucci, Jose Leibovich, and Diego Pizano, Colombia Coffee Sector Study (Bogotá: Universidad de los Andes–CEDE, 2002), published in Documentos CEDE.  
<sup>5</sup> Lady Gomez, Coffee Annual: Colombia (Bogotá: USDA Foreign Agricultural Service, 20 May 2024), Report No. CO2024-0007, approved by Rachel Nelson.  
<sup>6</sup> Gaspar Nolte, Coffee Annual: Peru (Lima: USDA Foreign Agricultural Service, 20 May 2024), Report No. PE2024-0008, approved by Zeke Bryant.

## Central America / Caribbean

### Honduras (4<sup>th</sup>)

Honduras concentrates its production in six official coffee-growing regions: Copán, Montecillos (La Paz and Intibucá), Opalaca (Lempira and Intibucá), Comayagua (Comayagua Department), El Paraíso (El Paraíso Department), and Agalta (Olancho).<sup>78</sup>

#### Arabica Main Growing Regions (Honduras)



<sup>7</sup> Juan Fiallos, Coffee Annual: Honduras (Tegucigalpa: USDA Foreign Agricultural Service, 8 May 2025), Report No. HO2025, approved by Jason Wrobel.

<sup>8</sup> Instituto Hondureño del Café (IHCAFE), Regiones cafetaleras de Honduras (Tegucigalpa: IHCAFE, n.d.), accessed 16 June 2025, <https://www.ihcafe.hn/regiones-cafetalaras/>.



## Guatemala (5<sup>th</sup>)

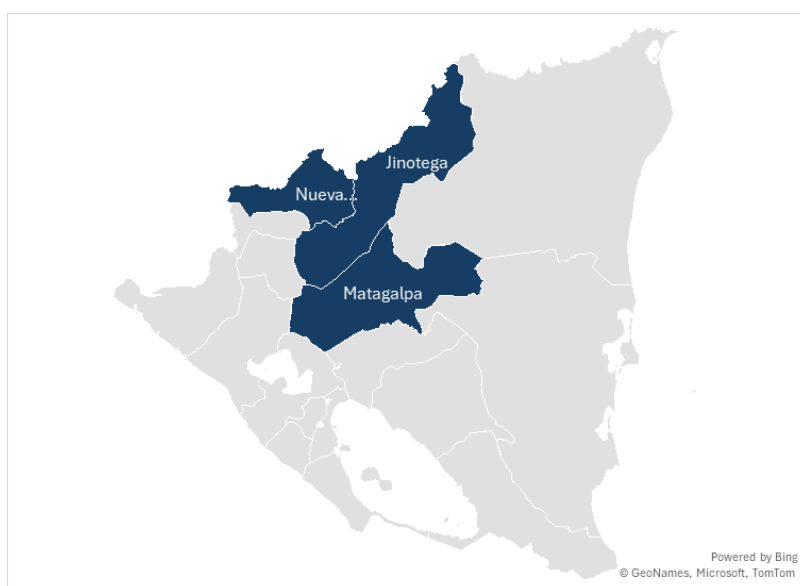
Finding official statistics from the USDA regarding the amount of Arabica produced in each region of Guatemala was difficult, especially as the country seems to put an emphasis on each regions distinctive type rather than a singular category. Nonetheless, the following regions seem to play a major role in the industry<sup>9</sup>.

### Arabica Main Growing Regions (Nicaragua)



## Nicaragua (8<sup>th</sup>)

### Arabica Main Growing Regions (Nicaragua)



More than 85 percent of the arabica coffee farms are in North Central Nicaragua in departments of Jinotega, Matagalpa, and Nueva Segovia, with a range of altitude from 365 to 1,500 meters above sea level<sup>10</sup>.

<sup>9</sup> Asociación Nacional del Café (Anacafé), Memoria de Labores 2023–2024 (Guatemala: Anacafé, 2024), accessed 17 June 2025, <https://www.anacafe.org/uploads/file/34527e51f4714908952a90c89ce0e551/Memoria-Labores-Anacafe-2024.pdf>.

<sup>10</sup> Coffee Annual: Nicaragua (Managua: USDA Foreign Agricultural Service, 16 May 2025), Report No. NU2025-0004, approved by Zeke Bryant.

## Mexico (7<sup>th</sup>)

### Arabica Main Growing Regions (Mexico)



Chiapas [38%], Veracruz [26%], Puebla [18%], and Oaxaca [9%] account for roughly 92% of coffee production in Mexico, although including production of robusta.<sup>11</sup>

## Africa

### Ethiopia (3<sup>rd</sup>)

Ethiopia is the third-largest producer and exporter of Arabica coffee, behind Brazil and Colombia<sup>12</sup> with the entire of its coffee production consisting of Arabica<sup>13</sup>. Ethiopia's main coffee production is located within Oromia [69%], and the former SNNPR area [29%]<sup>14</sup>

### Arabica Main Growing Regions (Ethiopia)<sup>10</sup>

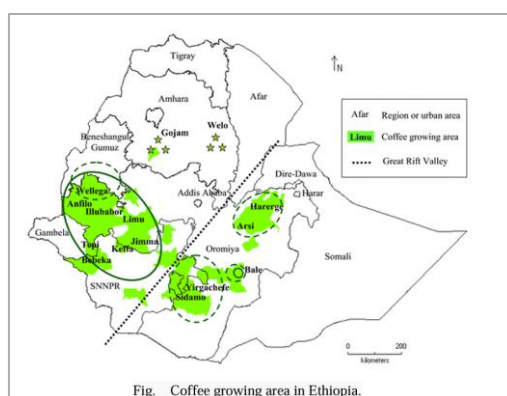


Fig. Coffee growing area in Ethiopia.

<sup>11</sup> Adriana Otero Arnaiz, Coffee Annual: Mexico (Mexico City: USDA Foreign Agricultural Service, 17 May 2024), Report No. MX2024-0025, approved by Alexander Chinh.

<sup>12</sup> FAS Staff, Coffee Annual: Ethiopia (Addis Ababa: USDA Foreign Agricultural Service, 3 June 2025), Report No. ET2025-0014, approved by Justina Torry.

<sup>13</sup> Abu Tefera, Coffee Annual: Ethiopia (Addis Ababa: USDA Foreign Agricultural Service, n.d.), approved by Justina Torry.

<sup>14</sup> Efrem Asfaw, "The Review Article: Ethiopian Coffee Production Trend Analysis: Comparison with Major Coffee Producing Countries of the World," International Journal of Social Sciences and Humanities Invention 11 (August 2024): 8234–8237, <https://doi.org/10.18535/ijsshi/v11i08.01>.

## Asia

### China (9<sup>th</sup>)

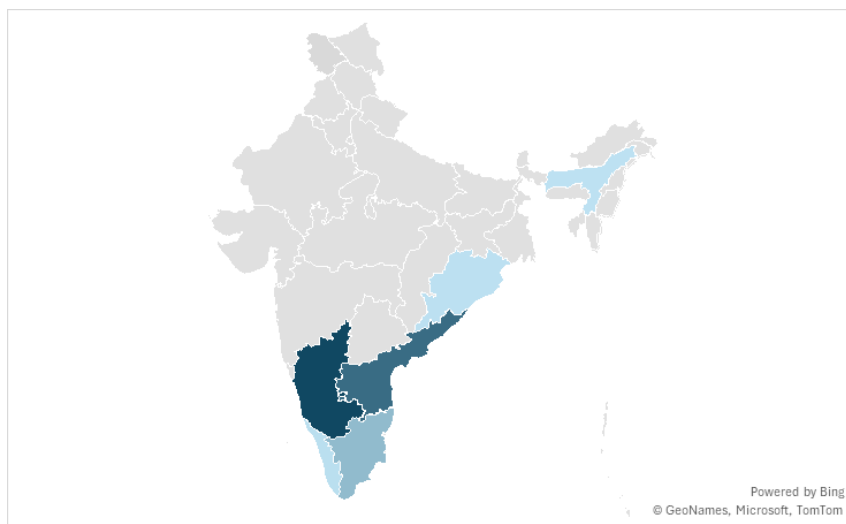
The Yunnan Province's plateaus (YN), comprises 98% of China's total coffee output in both cultivation area and yield<sup>15</sup>, with production centered in the prefectures of Baoshan, Dehong, Pu'er, and Lincang<sup>1</sup>.

#### Arabica Main Growing Regions (China)



### India (10<sup>th</sup>)

Roughly 95% of Indian arabica is grown in Karnataka [47%], Andhra Pradesh [36%], and Tamil Nadu [13%], in the southern part of the country<sup>16</sup>.



<sup>15</sup> Y. Zhu, Y. Liu, Z. Chen, et al., "Assessing the climate change impacts on Coffea arabica cultivation regions in China," Theoretical and Applied Climatology 155 (August 2024): 7773–7791, <https://doi.org/10.1007/s00704-024-05077-4>.

<sup>1</sup> United States Department of Agriculture Foreign Agricultural Service, Coffee: World Markets and Trade (Washington, DC: USDA FAS, December 2024).

<sup>16</sup> Dhruv Sood and Mariano J. Beillard, Coffee Annual – 2023: India (New Delhi: USDA Foreign Agricultural Service, 12 May 2023), Report No. IN2023-0033, approved by Mariano Beillard.

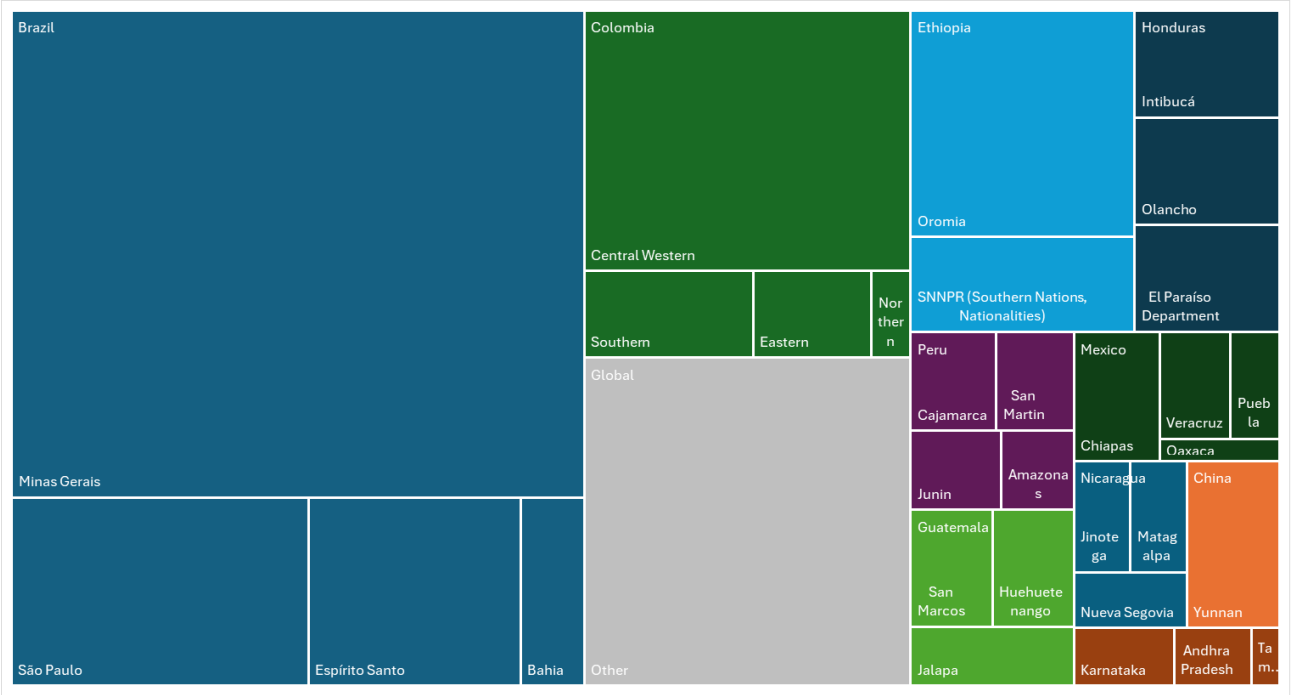
## Summary

| <i>Country</i>                | <i>Region/Province</i> | <i>National Production (%)</i> | <i>Global Production (%)</i> |
|-------------------------------|------------------------|--------------------------------|------------------------------|
| <b><i>Brazil</i></b>          |                        |                                | <b>46,40%</b>                |
|                               | Minas Gerais           | 70%                            | 32,48%                       |
|                               | São Paulo              | 14%                            | 6,50%                        |
|                               | Espírito Santo         | 10%                            | 4,64%                        |
|                               | Bahia                  | 3%                             | 1,39%                        |
|                               | Other                  | 3%                             | 1,39%                        |
| <b><i>Colombia</i></b>        |                        |                                | <b>13,18%</b>                |
|                               | Central Western        | 75%                            | 9,89%                        |
|                               | Southern               | 13%                            | 1,71%                        |
|                               | Eastern                | 9%                             | 1,19%                        |
|                               | Northern               | 3%                             | 0,40%                        |
| <b><i>Ethiopia</i></b>        |                        |                                | <b>8,54%</b>                 |
|                               | Oromia                 | 69%                            | 5,90%                        |
|                               | SNNPR                  | 29%                            | 2,48%                        |
|                               | Other                  | 2%                             | 0,17%                        |
| <b><i>Honduras</i></b>        |                        |                                | <b>5,42%</b>                 |
|                               | Intibucá               | 33%                            | 1,81%                        |
|                               | Olancho                | 33%                            | 1,81%                        |
|                               | El Paraíso             | 33%                            | 1,81%                        |
|                               | Department             |                                |                              |
| <b><i>Peru</i></b>            |                        |                                | <b>4,45%</b>                 |
|                               | Cajamarca              | 22%                            | 0,98%                        |
|                               | San Martin             | 20%                            | 0,89%                        |
|                               | Junin                  | 19%                            | 0,84%                        |
|                               | Amazonas               | 15%                            | 0,67%                        |
|                               | Other                  | 24%                            | 1,07%                        |
| <b><i>Mexico</i></b>          |                        |                                | <b>3,40%</b>                 |
|                               | Chiapas                | 38%                            | 1,29%                        |
|                               | Vecracruz              | 26%                            | 0,88%                        |
|                               | Puebla                 | 18%                            | 0,61%                        |
|                               | Oaxaca                 | 9%                             | 0,31%                        |
|                               | Other                  | 9%                             | 0,31%                        |
| <b><i>Guatemala</i></b>       |                        |                                | <b>3,37%</b>                 |
|                               | San Marcos             | 33%                            | 1,11%                        |
|                               | Huehuetenango          | 33%                            | 1,11%                        |
|                               | Jalapa                 | 33%                            | 1,11%                        |
| <b><i>Nicaragua</i></b>       |                        |                                | <b>2,58%</b>                 |
|                               | Jinotega               | 28%                            | 0,73%                        |
|                               | Matagalpa              | 28%                            | 0,73%                        |
|                               | Nueva Segovia          | 28%                            | 0,73%                        |
|                               | Other                  | 16%                            | 0,41%                        |
| <b><i>China</i></b>           |                        |                                | <b>1,84%</b>                 |
|                               | Yunnan                 | 98%                            | 1,80%                        |
|                               | Other                  | 2%                             | 0,04%                        |
| <b><i>India</i></b>           |                        |                                | <b>1,43%</b>                 |
|                               | Karnatka               | 47%                            | 0,67%                        |
|                               | Pradesh                | 36%                            | 0,52%                        |
|                               | Tamil Nadu             | 13%                            | 0,19%                        |
|                               | Other                  | 4%                             | 0,06%                        |
| <b><i>Other Countries</i></b> |                        |                                | <b>9%</b>                    |

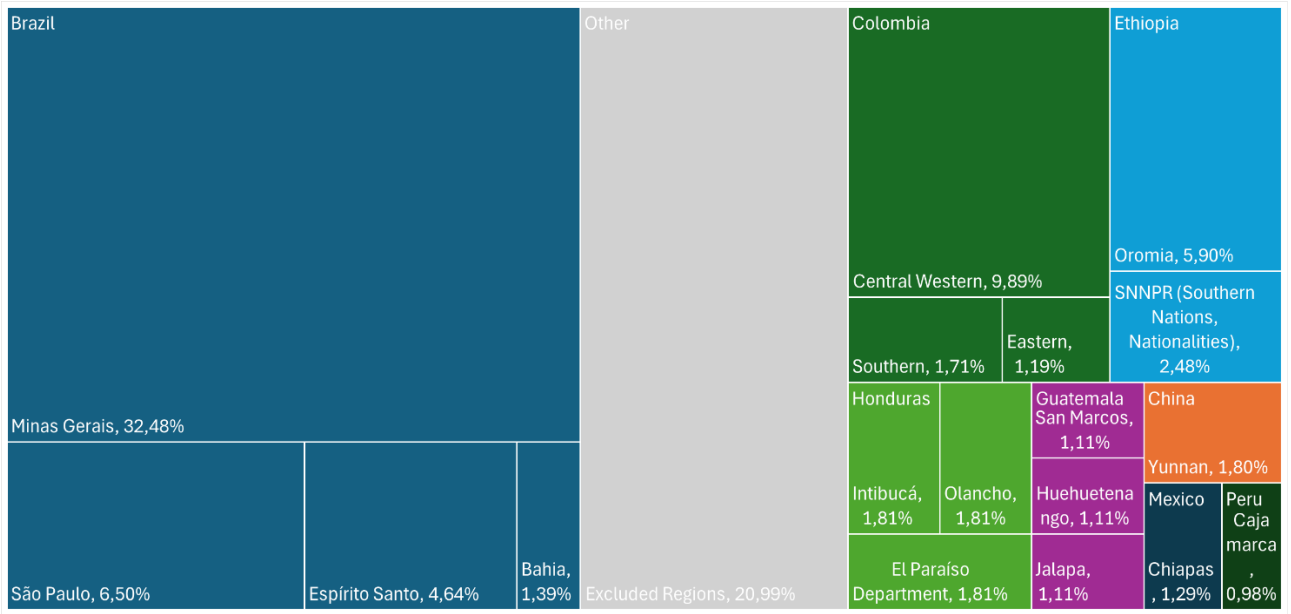


| COUNTRY          | REGION                | % GLOBAL PRODUCTION |
|------------------|-----------------------|---------------------|
| <b>BRAZIL</b>    | Minas Gerais          | <b>32,48%</b>       |
| <b>COLOMBIA</b>  | Central Western       | <b>9,89%</b>        |
| <b>BRAZIL</b>    | São Paulo             | <b>6,50%</b>        |
| <b>ETHIOPIA</b>  | Oromia                | <b>5,90%</b>        |
| <b>BRAZIL</b>    | Espírito Santo        | <b>4,64%</b>        |
| <b>ETHIOPIA</b>  | Former SNNPR          | <b>2,48%</b>        |
| <b>HONDURAS</b>  | Intibucá              | <b>1,81%</b>        |
| <b>HONDURAS</b>  | Olancho               | <b>1,81%</b>        |
| <b>HONDURAS</b>  | El Paraíso Department | <b>1,81%</b>        |
| <b>CHINA</b>     | Yunnan                | <b>1,80%</b>        |
| <b>COLOMBIA</b>  | Southern              | <b>1,71%</b>        |
| <b>BRAZIL</b>    | Bahia                 | <b>1,39%</b>        |
| <b>MEXICO</b>    | Chiapas               | <b>1,29%</b>        |
| <b>COLOMBIA</b>  | Eastern               | <b>1,19%</b>        |
| <b>GUATEMALA</b> | San Marcos            | <b>1,11%</b>        |
| <b>GUATEMALA</b> | Huehuetenango         | <b>1,11%</b>        |
| <b>GUATEMALA</b> | Jalapa                | <b>1,11%</b>        |
| <b>PERU</b>      | Cajamarca             | <b>0,98%</b>        |
| BRAZIL           | Other                 | 1,39%               |
| PERU             | Other                 | 1,07%               |
| PERU             | San Martin            | 0,89%               |
| MEXICO           | Vecracruz             | 0,88%               |
| PERU             | Junin                 | 0,84%               |
| NICARAGUA        | Jinotega              | 0,73%               |
| NICARAGUA        | Matagalpa             | 0,73%               |
| NICARAGUA        | Nueva Segovia         | 0,73%               |
| PERU             | Amazonas              | 0,67%               |
| INDIA            | Karnatka              | 0,67%               |
| MEXICO           | Puebla                | 0,61%               |
| INDIA            | Andhra Pradesh        | 0,52%               |
| NICARAGUA        | Other                 | 0,41%               |
| COLOMBIA         | Northern              | 0,40%               |
| MEXICO           | Oaxaca                | 0,31%               |
| MEXICO           | Other                 | 0,31%               |
| INDIA            | Tamil Nadu            | 0,19%               |
| ETHIOPIA         | Other                 | 0,17%               |
| INDIA            | Other                 | 0,06%               |
| CHINA            | Other                 | 0,04%               |
| GLOBAL           | Other                 | 9%                  |

Regional Treemap



Regional treemap Excluding, minor areas



Based on the following, multiple datapoints ought to be collected from those areas that play most role in the global production of arabica.

## Coordinates (Nasapower)

NASA POWER's API provides weather variables at the scale of large climate zones ( $\sim 55 \times 69$  km), with grid cells approaching  $\sim 55 \times 55$  km near the equator. Because Arabica coffee cultivation depends heavily on microclimates, this level of resolution may not capture the fine-scale variations essential for suitability assessments. Therefore, it is important to confirm that the chosen locations for data extraction contain substantial coffee farming areas to ensure the relevance of the climate data to actual plantations<sup>17</sup>. The following can be achieved via two complementary methods: climate & elevation modelling, and visual confirmation.

## Climate & Elevation Modelling

NASA's Worldview can provide valuable visual information about temperature, elevation, and vegetation, helping to pinpoint regions where Arabica coffee is most likely to thrive. Arabica is a delicate crop with strict environmental needs, particularly concerning temperature, elevation, rainfall, and shade. Ideally, Arabica grows best between 18°C and 23°C, though it can still perform reasonably well up to 24–25°C with intensive management<sup>18</sup>. It generally flourishes at elevations above 1,000 meters, with a minimum suitable range often considered to be 700 to 900 meters<sup>19</sup>.

Rainfall and shading present more complex challenges. Both too little and too much moisture during different growth stages can lead to issues like fruit malformation or fungal outbreaks<sup>20</sup>. While shade benefits the plant by improving its physiological performance and reducing temperature extremes<sup>21</sup>, commercial production, especially in areas like Brazil, often favors full-sun systems for higher yields<sup>22</sup>. This is despite potential trade-offs in plant health and environmental sustainability. Therefore, while rainfall and shading are crucial metrics, they should not be the sole determinants of suitable Arabica growing regions.



<sup>17</sup> NASA Langley Research Center, NASA POWER Tutorials (Hampton, VA: NASA Langley Research Center, n.d.), accessed 18 June 2025, <https://power.larc.nasa.gov/docs/tutorials/>.

<sup>18</sup> G. de Souza Rolim, L. E. de Oliveira Aparecido, P. S. de Souza, et al., "Climate and Natural Quality of *Coffea arabica* L. Drink," Theoretical and Applied Climatology 141 (July 2020): 87–98, <https://doi.org/10.1007/s00704-020-03117-3>.

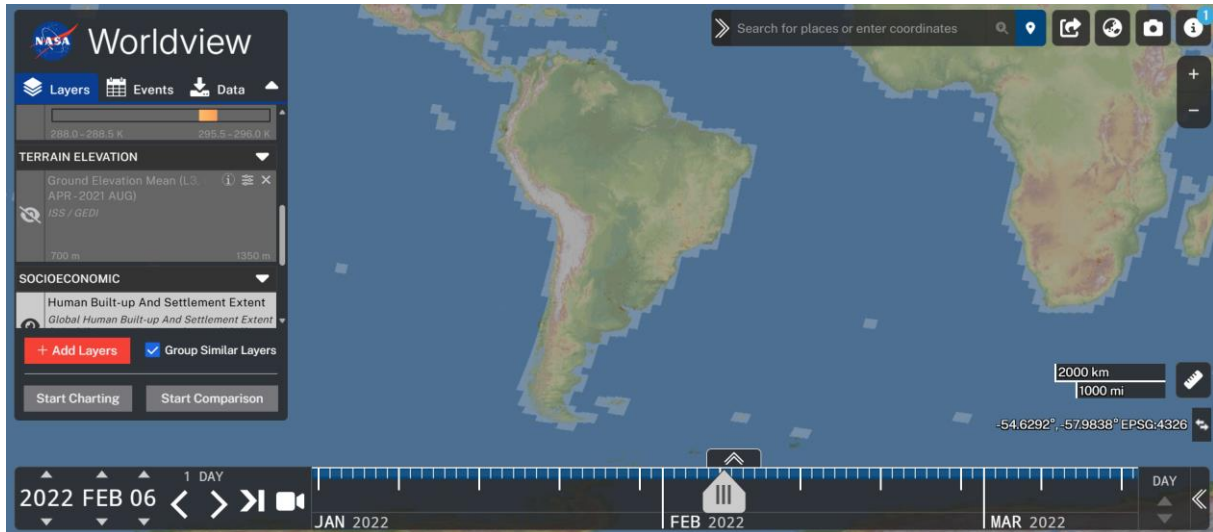
<sup>19</sup> Dwi Nugroho, Panjisakti Basunanda, and Suyadi Mw, "Physical Bean Quality of Arabica Coffee (*Coffea arabica*) at High and Medium Altitude," Pelita Perkebunan 32, no. 3 (2016): n.p.

<sup>20</sup> Jarrod Kath, Vivekananda Mittahalli Byrareddy, Shahbaz Mushtaq, Alessandro Craparo, and Mario Porcel, "Temperature and Rainfall Impacts on Robusta Coffee Bean Characteristics," Climate Risk Management 32 (2021): 100281, <https://doi.org/10.1016/j.crm.2021.100281>.

<sup>21</sup> R. G. Muschler, "Shade Improves Coffee Quality in a Sub-Optimal Coffee-Zone of Costa Rica," Agroforestry Systems 51 (January 2001): 131–139, <https://doi.org/10.1023/A:1010603320653>.

<sup>22</sup> A. Koutouleas, T. Sarzynski, B. Bertrand, et al., "Shade Effects on Yield Across Different *Coffea arabica* Cultivars — How Much Is Too Much? A Meta-Analysis," Agronomy for Sustainable Development 42 (2022): 55, <https://doi.org/10.1007/s13593-022-00788-2>.

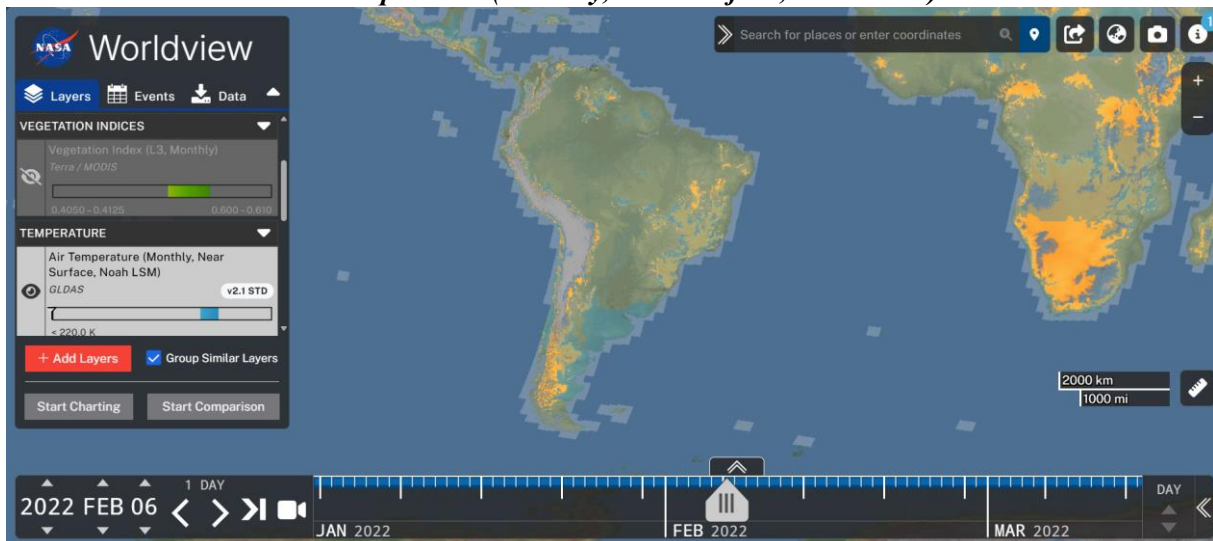
## Nasa Worldview



## Ground Elevation Mean (L3, v2, 2019 APR - 2021 AUG)



## Air Temperature (Monthly, Near Surface, Noah LSM)



## Vegetation Index (L3, Monthly)





**Soil Moisture (Normalized Polarization Difference, Day)**



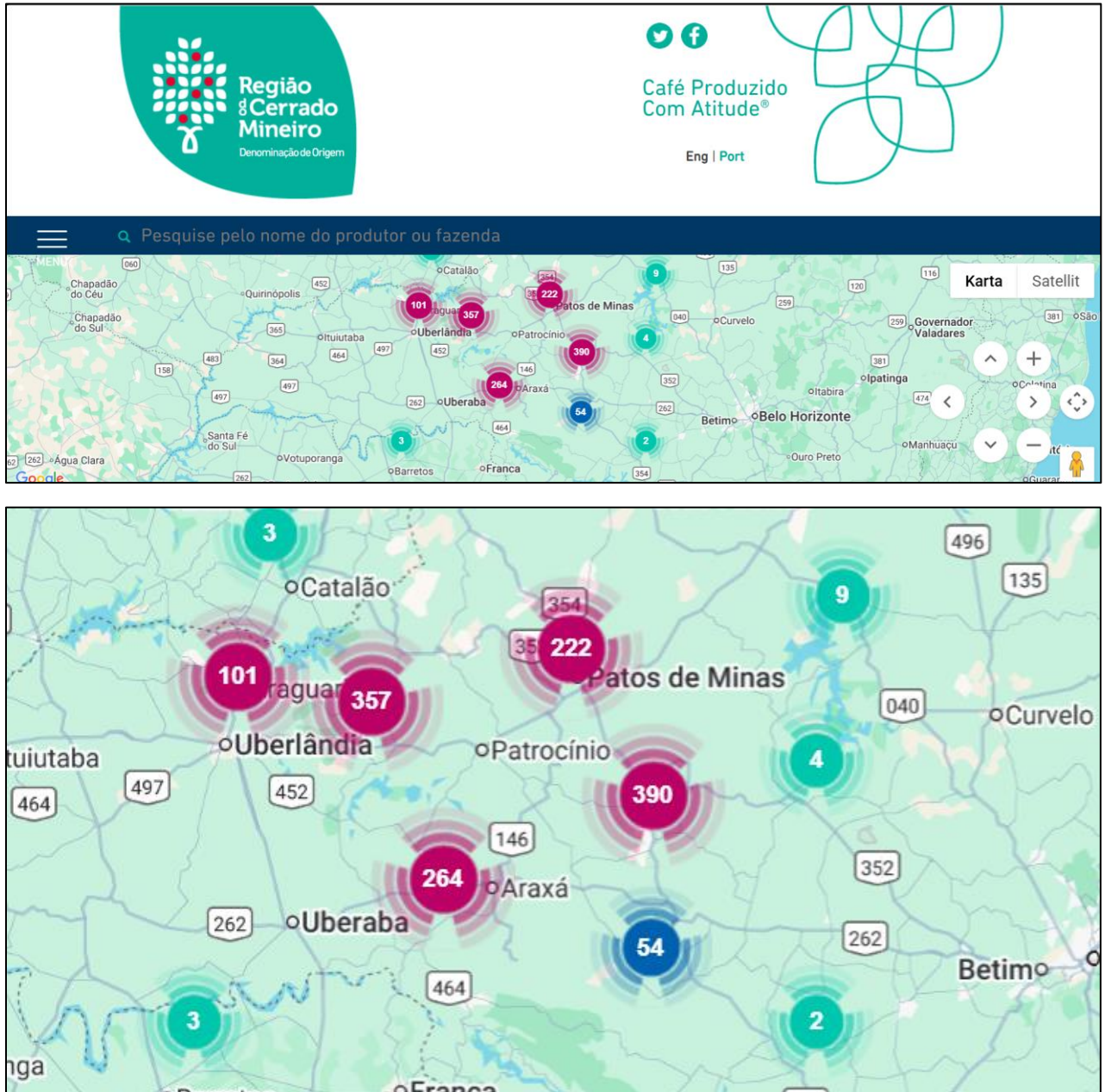
| DEPTH/SEASON          | % VOLUMETRIC MOISTURE | G OF WATER PER CM <sup>3</sup> OF SOIL. |
|-----------------------|-----------------------|-----------------------------------------|
| 10 CM DEEP/DRY SEASON | 14.1%                 | 0.141 g/cm <sup>3</sup>                 |
| 10 CM DEEP/WET SEASON | 27.8%                 | 0.278 g/cm <sup>3</sup>                 |
| 20 CM DEEP/DRY SEASON | 48.5%                 | 0.485 g/cm <sup>3</sup>                 |
| 20 CM DEEP/WET SEASON | 62.3%                 | 0.623 g/cm <sup>3</sup>                 |

<sup>23</sup>Note AMSR2 only looks at surface level (~2cm), but the following table provides an understanding of what g of water per cm<sup>3</sup> of soil is typically common for Arabica.

<sup>23</sup> Flórez, L., Cartagena, J. and Ramírez Builes, V., 2018. Soil humidity and evapotranspiration under three coffee (Coffea arabica L.) planting densities at Naranjal experimental station (Chinchiná, Caldas, Colombia). Acta Agronómica, 67(3), pp.402–413. <https://doi.org/10.15446/acag.v67n3.67377>.

## Visual Confirmation

The Federation of Coffee Growers of the Cerrado (Federação dos Cafeicultores do Cerrado) provides detailed information on coffee farms in the Cerrado Mineiro region of Minas Gerais<sup>24</sup>. This data is instrumental for understanding how coffee plantations appear in satellite imagery<sup>25</sup>, which is especially valuable when extracting information from regions where detailed farm-level data is lacking. While applying machine learning techniques to satellite images to identify coffee plantations could significantly aid this process, such an approach is beyond the scope of the current project.

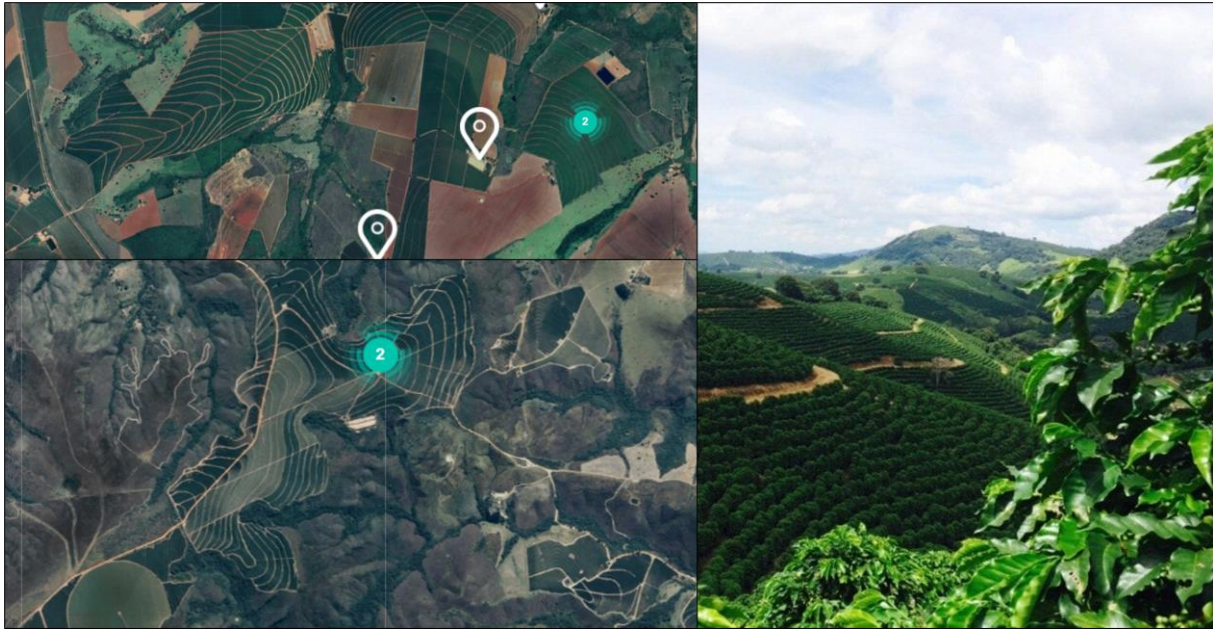


<sup>24</sup> Federação dos Cafeicultores do Cerrado, Denominação de Origem – Região do Cerrado Mineiro (Tegucigalpa: Federação dos Cafeicultores do Cerrado, n.d.), accessed in 18 de junho de 2025, <https://cerradomineiro.org/index.php?pg=denominacaodeorigem#group1>.

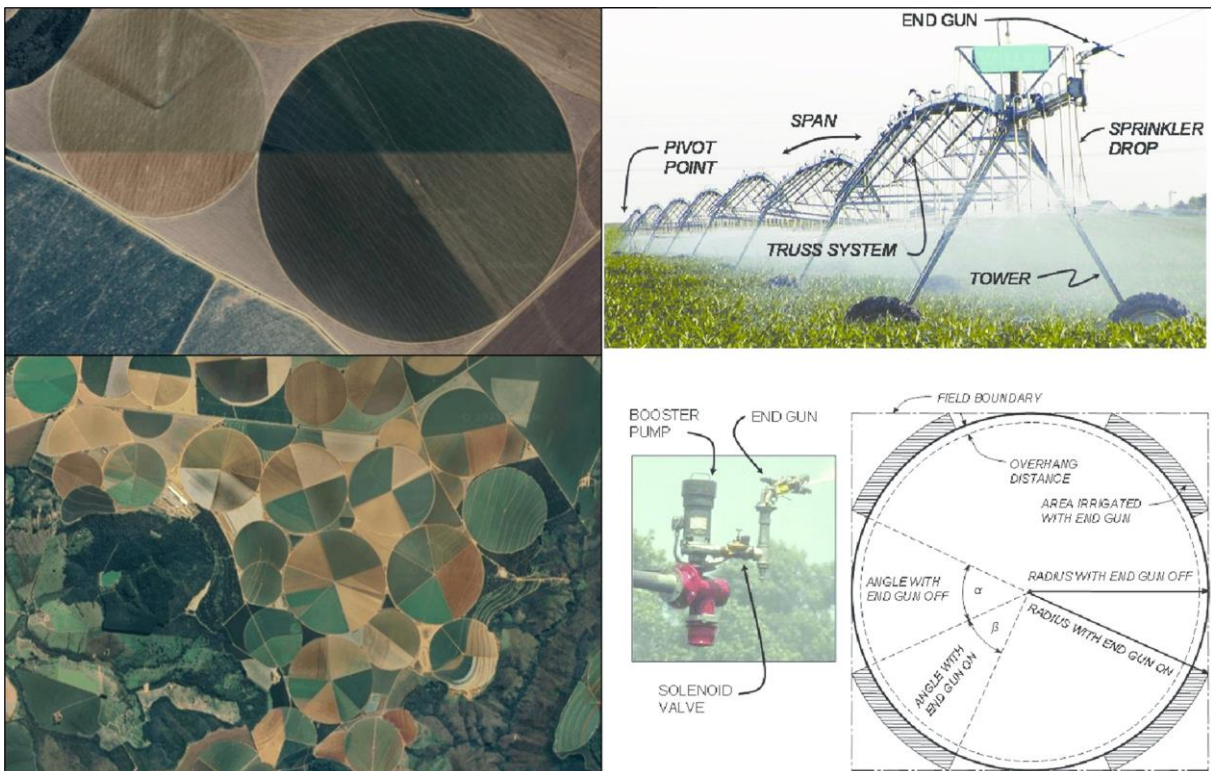
<sup>25</sup> Federação dos Cafeicultores do Cerrado, Mapa das Fazendas – Região do Cerrado Mineiro (Tegucigalpa? likely Brazil: Cerrado Mineiro, n.d.), accessed 18 June 2025, <https://www.cerradomineiro.org/index.php?pg=mapa>.  
cerradomineiro.org



### High likelihood of being a coffee plantation (Terraced Farm)



### Lower likelihood of being a coffee plantation (Center-Pivot Irrigation)



The center pivot irrigation system is a relatively recent method that uses a single sprinkler pipeline rotating around a central pivot<sup>26</sup>. It was introduced on a large scale for coffee irrigation in Brazil starting in the late 1990s<sup>27</sup>. Although center pivot systems are common for various types of crops, their presence alone is not a reliable indicator of coffee plantations in satellite imagery. Additionally, the

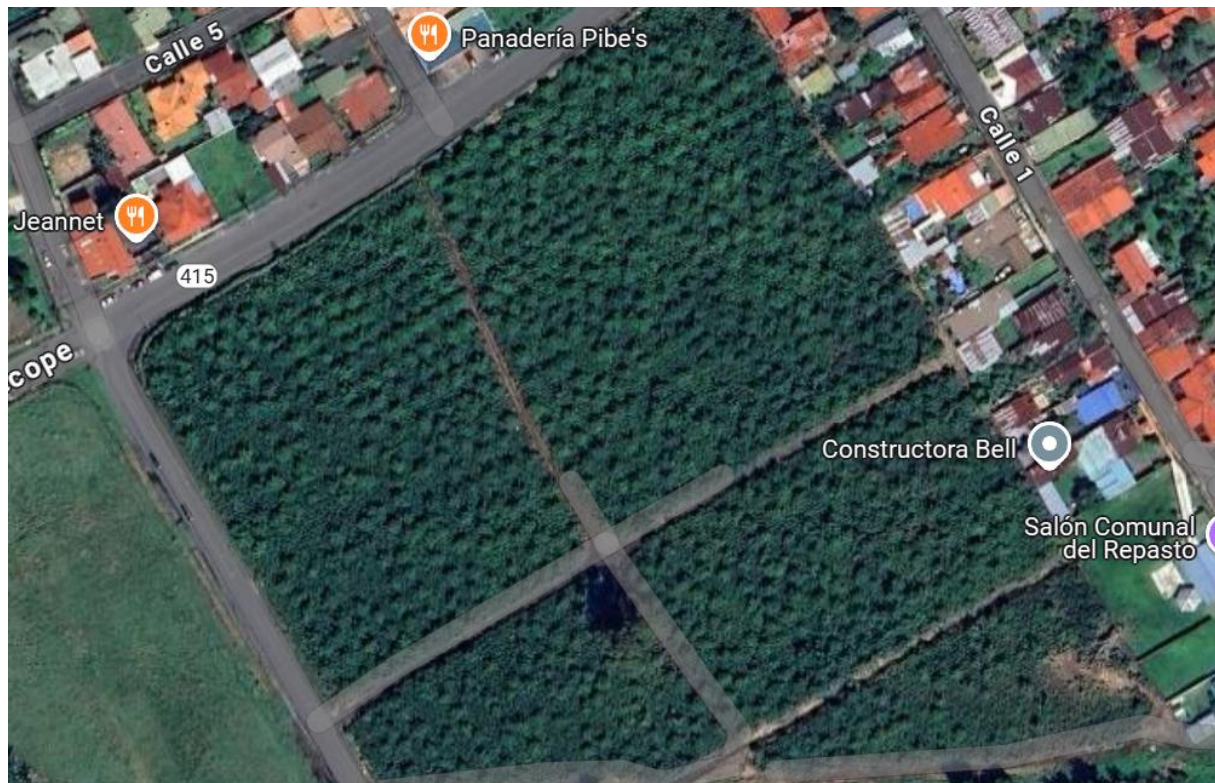
<sup>26</sup> Chapter 10: The Center Pivot Irrigation Systems, in Technical Handbook on Pressurized Irrigation Techniques (Rome: FAO, n.d.), accessed 18 June 2025, <https://www.fao.org/4/a1336e/a1336e10a.pdf>.

<sup>27</sup> Adriano Custódio, Edson Pozza, Anselmo Custódio, Paulo Souza, Luiz Lima, and Antônio Silva, "Effect of Center-Pivot Irrigation in the Rust and Brown Eye Spot of Coffee," *Plant Disease* 98 (July 2014): 943–947, <https://doi.org/10.1094/PDIS-07-13-0801-RE>.

high costs associated with this irrigation method make it less feasible in less mechanized or lower-income coffee-growing regions outside Brazil.<sup>28</sup>

### **High likelihood of being a coffee plantation (Mature AFS) \***

Despite dense vegetation making it harder to determine a plot of coffee via satellite, especially in rural areas, it also can aid in confirming whether the area in question is used to cultivate arabica. Attention should be drawn as to whether other types of fruit trees are grown in the area of interest.



*Mature AFS plot located in in Turrialba, Costa Rica (9.917000, -83.683000) used in a previous study about shading<sup>29</sup>.*

<sup>28</sup> Nelson Nic, "Understanding the Pros and Cons of Center Pivot Irrigation Before Making the Switch," Irrigation Europe, accessed 19 June 2025, <https://irrigationeurope.eu/en/understanding-the-pros-and-cons-of-center-pivot-irrigation-before-making-the-switch/>.

<sup>29</sup> R. G. Muschler, "Shade Improves Coffee Quality in a Sub-Optimal Coffee-Zone of Costa Rica," *Agroforestry Systems* 51 (January 2001): 131–139, <https://doi.org/10.1023/A:1010603320653>.

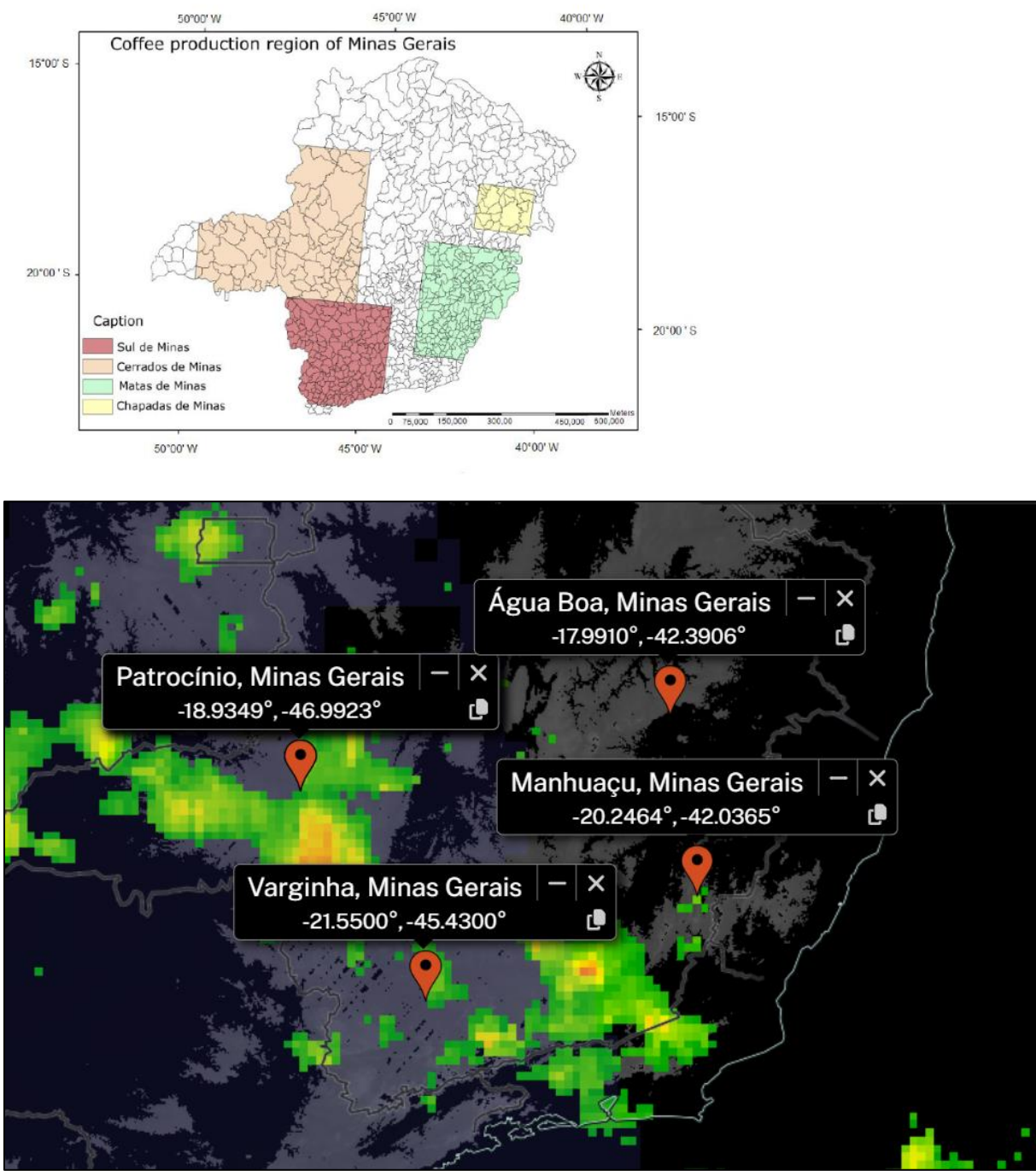


## Study Areas

| Location               | Region         | Country   | Latitude | Longitude |
|------------------------|----------------|-----------|----------|-----------|
| Patrocínio             | Minas Gerais   | Brazil    | -18.9349 | -46.9923  |
| Varginha               | Minas Gerais   | Brazil    | -21.5500 | -45.4300  |
| Manhuaçu               | Minas Gerais   | Brazil    | -20.2464 | -42.0365  |
| Água Boa               | Minas Gerais   | Brazil    | -17.9910 | -42.3906  |
| Campinas               | São Paulo      | Brazil    | -22.7884 | -47.0579  |
| Sao Jose Da Bela Vista | São Paulo      | Brazil    | -20.5333 | -47.5833  |
| Afonso Claudio         | Espirito Santo | Brazil    | -20.0421 | -40.9743  |
| Conceicao do Castelo   | Espirito Santo | Brazil    | -20.3633 | -41.2502  |
| Morelia                | Caquetá        | Colombia  | 1.3799   | -75.6736  |
| Andes                  | Antioquia      | Colombia  | 5.6959   | -75.8469  |
| Risaralda              | Santuario      | Colombia  | 5.0608   | -75.9548  |
| Timbio                 | Cauca          | Colombia  | 2.3433   | -76.6960  |
| Limu Kosa              | Oromia         | Ethiopia  | 7.9320   | 36.8249   |
| Bare                   | Oromia         | Ethiopia  | 8.2428   | 35.2119   |
| Aleto Wondo            | Former SSNPR   | Ethiopia  | 6.5759   | 38.4061   |
| Aleto Wondo            | Former SSNPR   | Ethiopia  | 6.8553   | 35.3337   |
| San Jose               | La Paza        | Honduras  | 14.2203  | -87.9285  |
| Danli                  | El Paraiso     | Honduras  | 13.9339  | -86.2882  |
| Campamento             | Olancho        | Honduras  | 14.5579  | -86.6347  |
| Sibinal                | San Marcos     | Guatemala | 15.1130  | -92.0892  |
| San Pedro Necta        | Huehuetenango  | Guatemala | 15.5633  | -91.7718  |
| Jalapa                 | Jalapa         | Guatemala | 14.7118  | -89.9874  |
| Jaen                   | Cajamarca      | Peru      | -5.6981  | -78.7938  |
| Puer City              | Yunnan         | China     | 22.6555  | 100.9535  |
| Las Maravillas         | Chiapas        | Mexico    | 15.1764  | -92.3310  |
|                        |                |           |          |           |

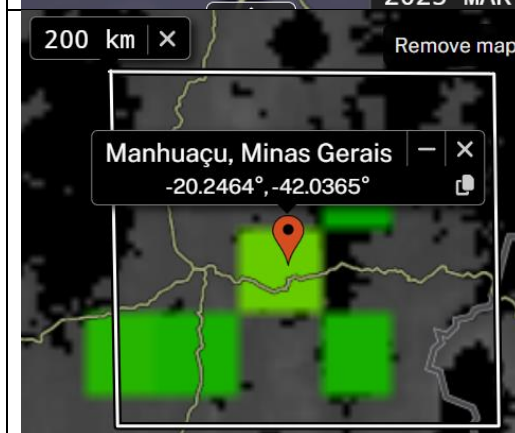
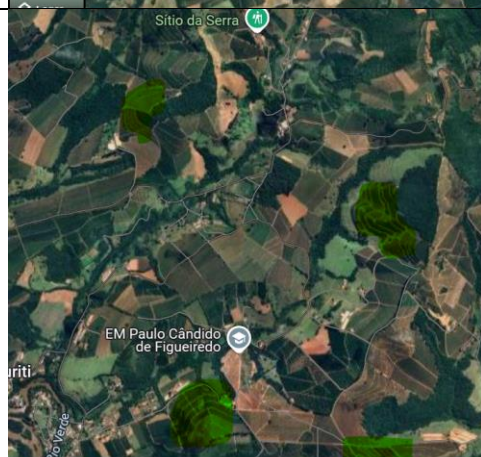
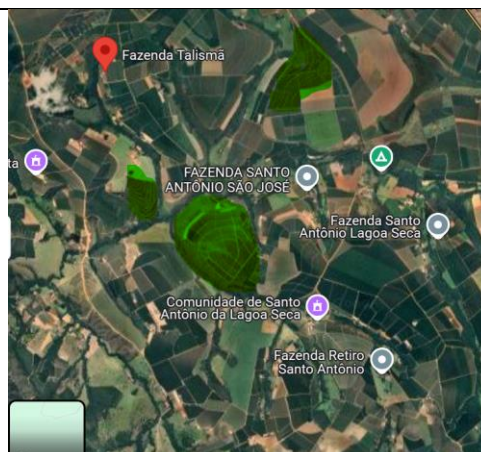
Minas Gerais - Brazil [32,48%]

Minas Gerais accounts for roughly 30% of global production on a yearly basis and thus includes multiple collection points, based on the main coffee production region within the state.<sup>30</sup>



| Overview | Visual Confirmation |
|----------|---------------------|
|----------|---------------------|

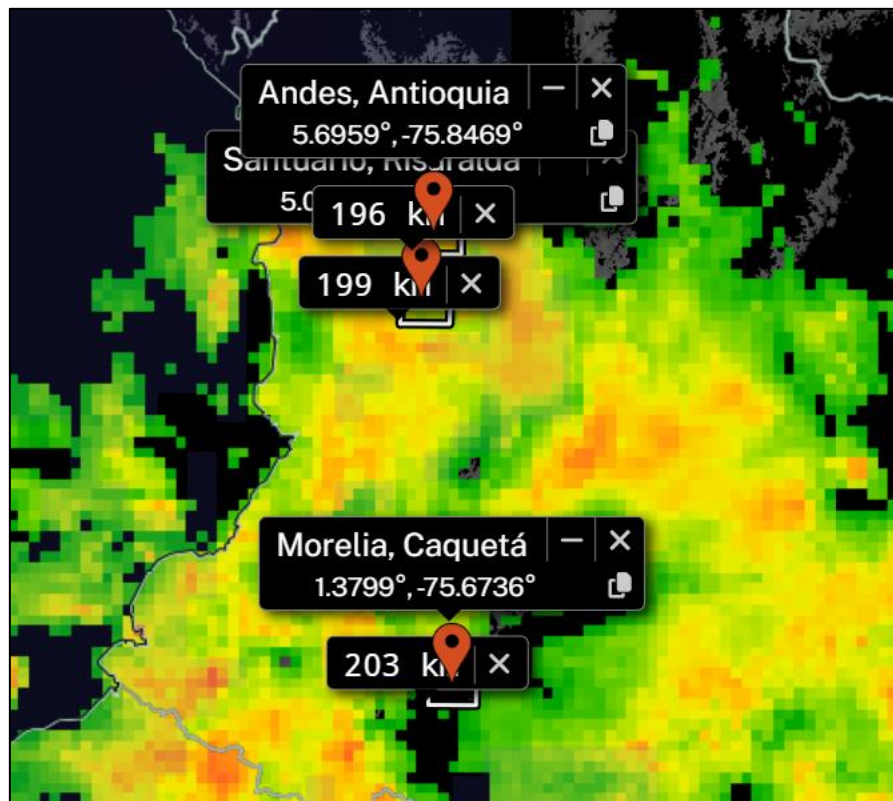
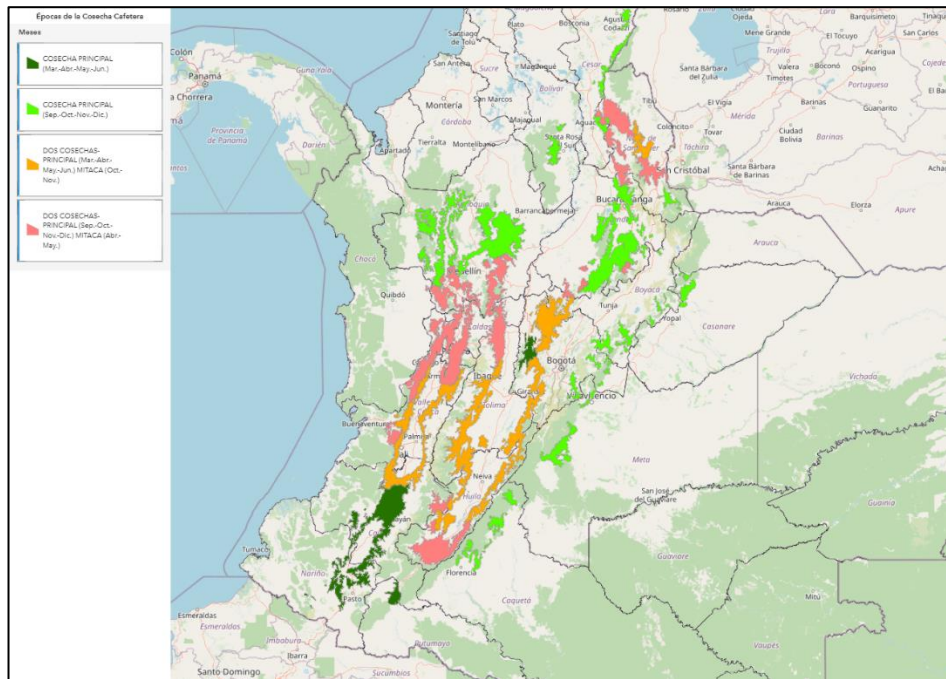
<sup>30</sup> Barbosa, J., Meira, F., Borém, F., Cirillo, M., Malta, M., Alvarenga, A., Helena, M. and Alves, H., 2 012. Coffee quality and its interactions with environmental factors in Minas Gerais, Brazil. Journal of Agricultural Science, 4(5), p.181. <https://doi.org/10.5539/jas.v4n5p181>



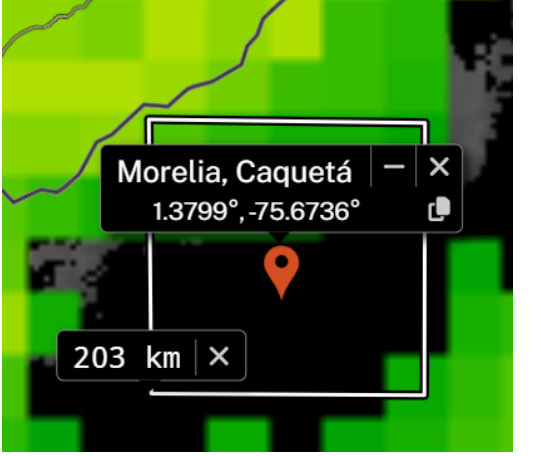
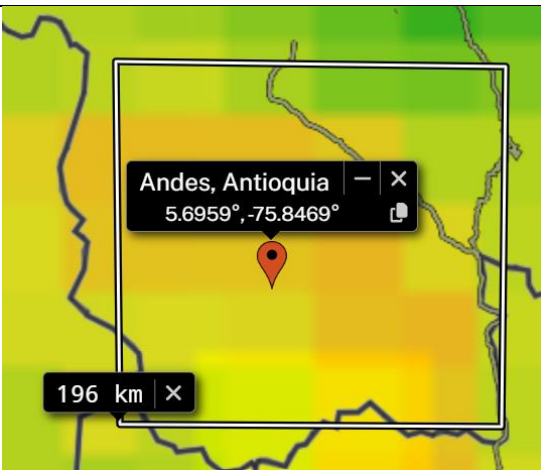
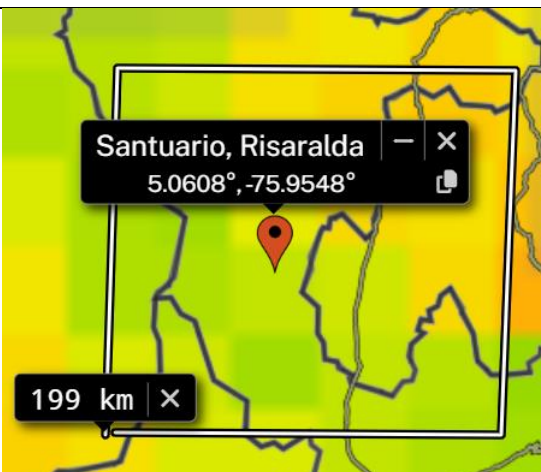


## Central Western Colombia [9,89%]

Due to the mountainous climate in Colombia compared to Minas Gerais, multiple data points were collected to account for potential microclimates forming between chains.



<sup>31</sup> Federación Nacional de Cafeteros de Colombia, Coffee Harvest Map (Bogotá: Federación Nacional de Cafeteros de Colombia, n.d.), accessed 3 July 2025, <https://federaciondefcafeteros.org/wp/coffee-harvest-map/?lang=en>.

| Overview                                                                                                                                                                                                                       | Visual Confirmation                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Map showing the location of Morelia, Caquetá at coordinates 1.3799°, -75.6736°. The map includes a scale bar indicating 203 km.</p>       | <p>32</p>  <p>Photograph showing a landscape view of Morelia, Caquetá, featuring rolling hills and a mountain range in the background.</p>  |
|  <p>Map showing the location of Andes, Antioquia at coordinates 5.6959°, -75.8469°. The map includes a scale bar indicating 196 km.</p>      |  <p>33</p> <p>Aerial photograph showing a landscape view of Andes, Antioquia, featuring terraced coffee fields and dense forest.</p>       |
|  <p>Map showing the location of Santuario, Risaralda at coordinates 5.0608°, -75.9548°. The map includes a scale bar indicating 199 km.</p> |  <p>Satellite image showing the location of Montes Tatama Specialty Coffee, featuring a small building and surrounding forested area.</p> |

<sup>32</sup> Wendell Rowan, Finca La Maria, photograph, June 2021, Google Maps, accessed 3 July 2025, <https://www.google.com/maps>.

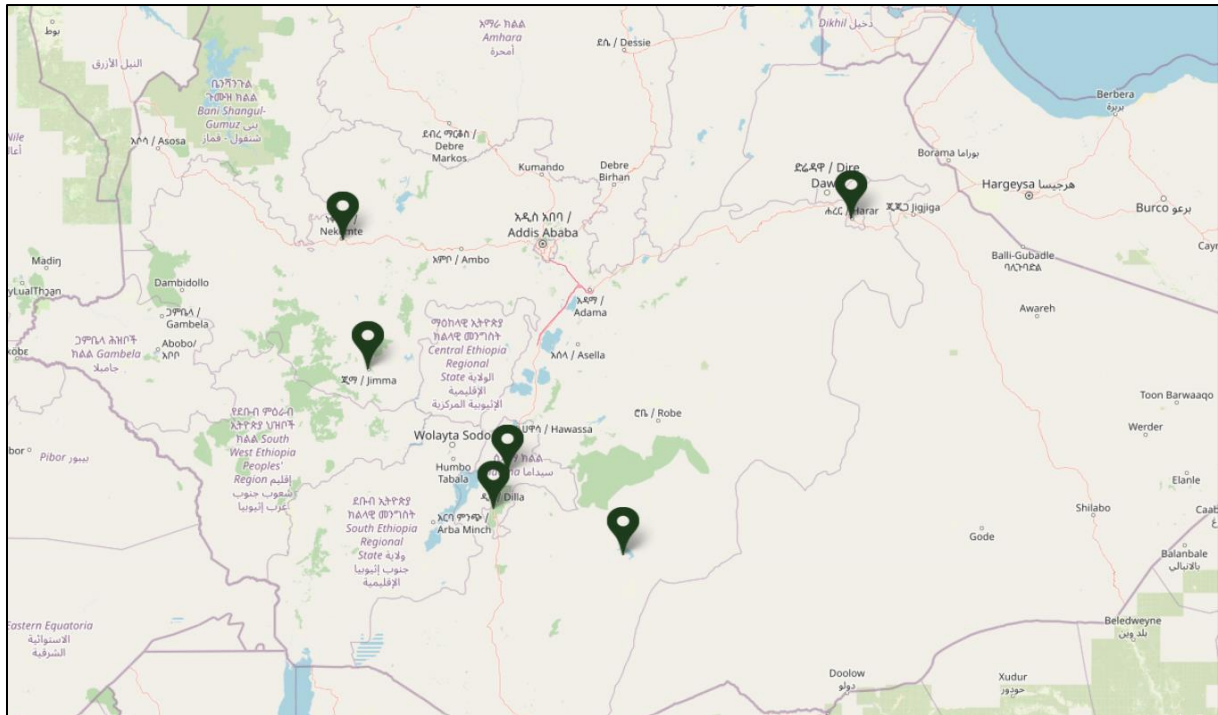
<sup>33</sup> Antioquia Coffee, “Our Story,” Antioquia Coffee, accessed 3 July 2025, <https://www.antioquiacooffee.shop/our-story>.



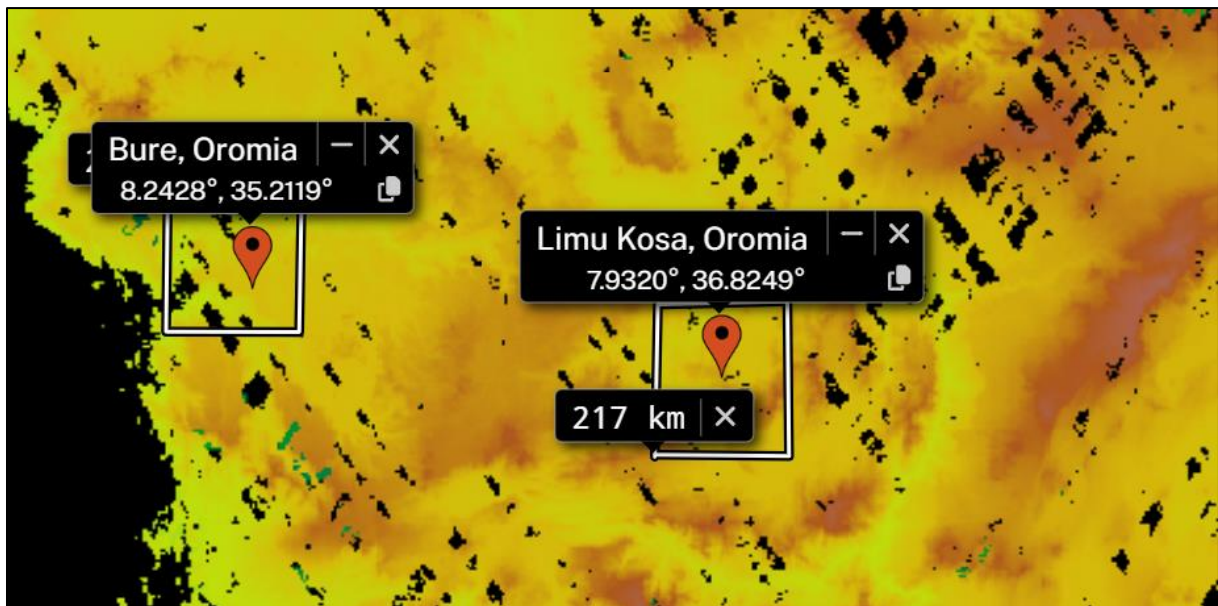
## São Paulo – Brazil [6,50%]

| Overview                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Visual Confirmation                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>A map showing the location of Campinas, São Paulo. The map is dark with yellow lines representing roads. A red location pin is placed on the map. A white box highlights the area around the pin. A black box with white text displays the location name and coordinates: "Campinas, Sao Paulo -22.7484°, -47.0579°". A scale bar indicates 252 km.</p>                              |  <p>A photograph showing a vast vineyard under a clear blue sky. The rows of grapevines stretch across a green field, with a few trees visible in the distance.</p>                                                                                                                     |
|  <p>A map showing the location of São José da Bela Vista, São Paulo. The map is dark with yellow lines representing roads. A red location pin is placed on the map. A white box highlights the area around the pin. A black box with white text displays the location name and coordinates: "São José da Bela Vista, São Paulo -20.5333°, -47.5833°". A scale bar indicates 201 km.</p> |  <p>A satellite image showing a rural landscape with green fields, brown patches, and a winding road. A location pin is placed on the map. Labels on the map include "Geleco Geleiras", "Distrito Industrial", and "Linha Refinaria &amp; Lavadora". A scale bar indicates 201 km.</p> |

## Oromia – Ethiopia [5,90%]

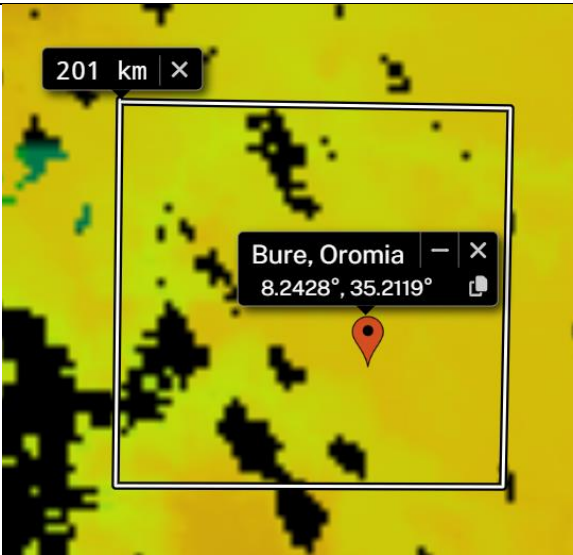
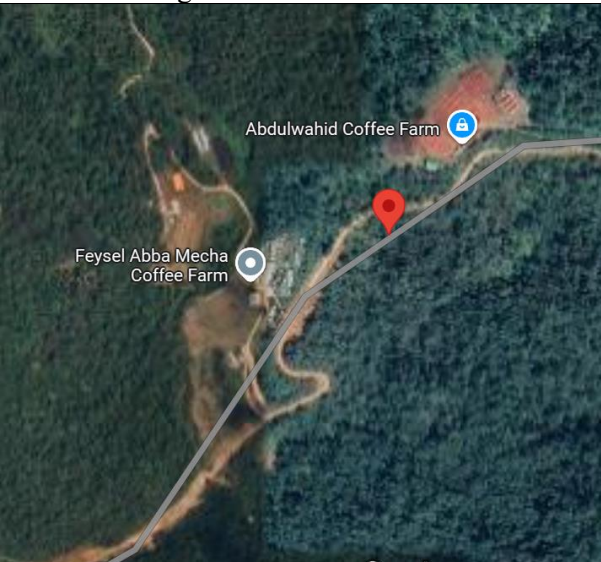


34

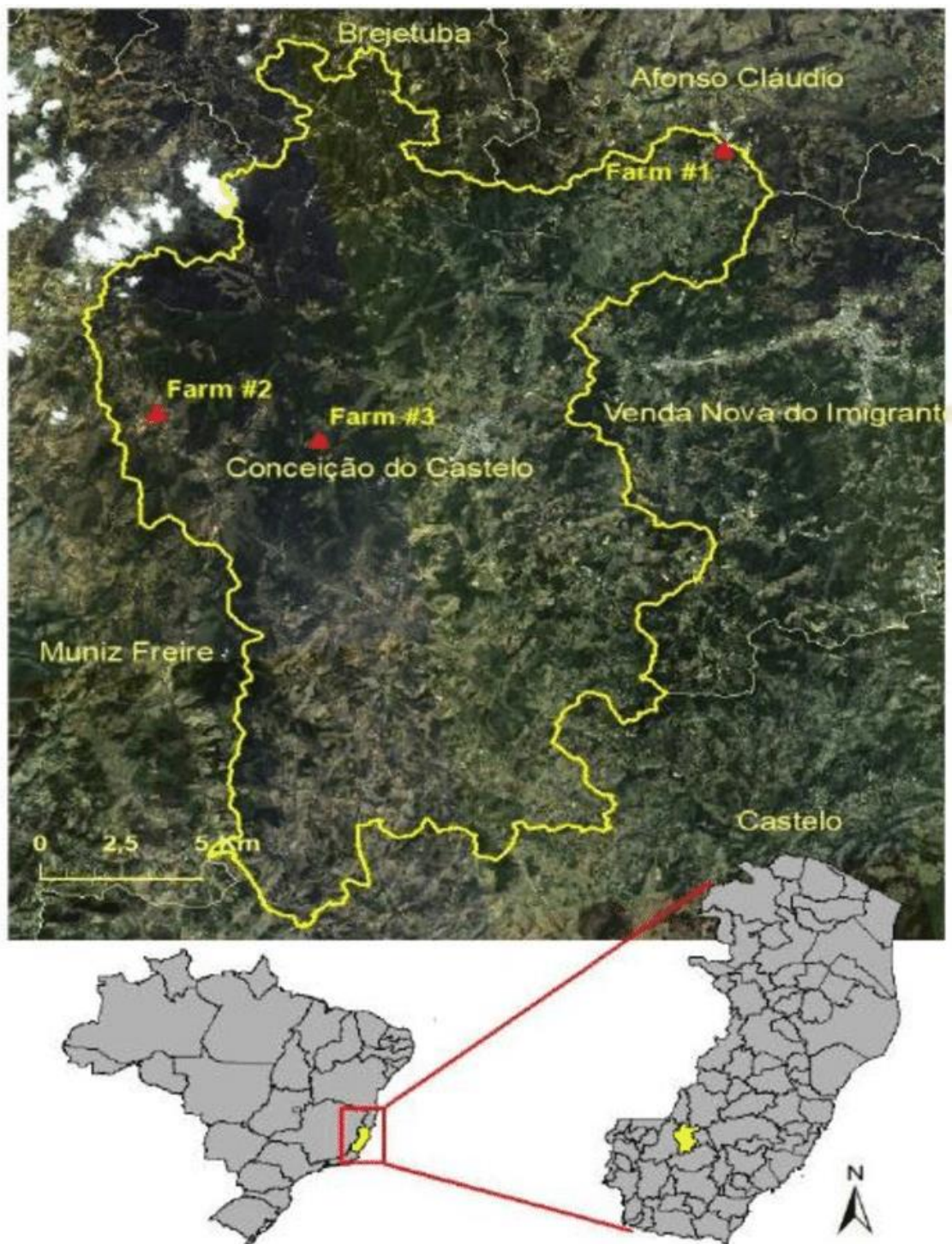


<sup>34</sup> Oromia Coffee Farmers' Cooperative Union, "Oromia," Oromia Coffee Union, accessed 3 July 2025, <https://www.oromiacoffeeunion.com/oromia>.



| Overview                                                                                                                                                                                                                                                                                                | Visual Confirmation                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Overview map showing the location of Limu Kosa, Oromia. The map displays a yellowish terrain with a red location pin. A scale bar indicates 217 km. The location is marked with coordinates 7.9320°, 36.8249°.</p> |  <p>Satellite image showing the Mamo Abdu Arabica coffee Farm. The farm is highlighted in green, indicating AFS shades. The surrounding area is forested.</p> <p>AFS shades in green</p>    |
|  <p>Overview map showing the location of Bure, Oromia. The map displays a yellowish terrain with a red location pin. A scale bar indicates 201 km. The location is marked with coordinates 8.2428°, 35.2119°.</p>     |  <p>Satellite image showing the Abdulwahid Coffee Farm and Feysel Abba Mecha Coffee Farm. The farms are highlighted in green, indicating AFS shades. The surrounding area is forested.</p> |

Espírito Santo – Brazil [4,64%]



35

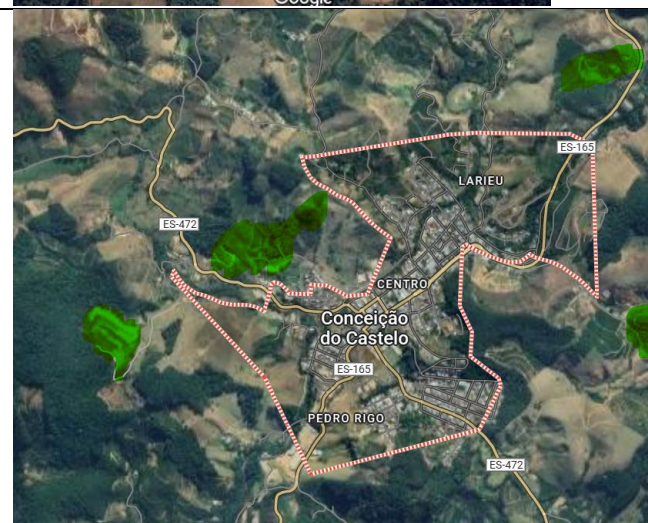
<sup>35</sup> Assessing the sustainability of Arabica coffee-growing householders in the Montanhas do Espírito Santo' geographic indication region, Brazil: an integrated case study - Scientific Figure on ResearchGate," ResearchGate, available from: [https://www.researchgate.net/figure/Map-of-the-location-of-coffee-farms-according-to-geographic-coordinates-altitude-and\\_fig1\\_363059248](https://www.researchgate.net/figure/Map-of-the-location-of-coffee-farms-according-to-geographic-coordinates-altitude-and_fig1_363059248) [accessed 3 Jul. 2025]



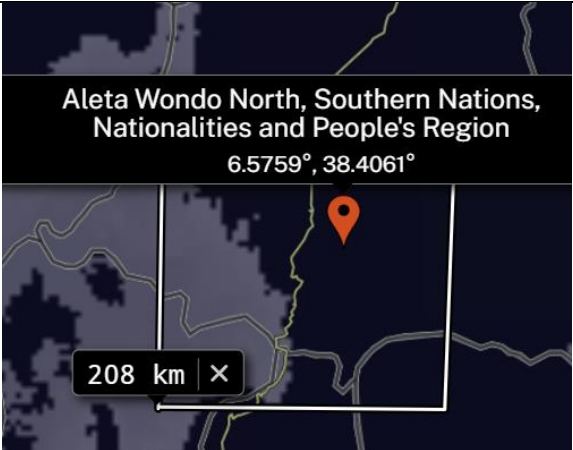
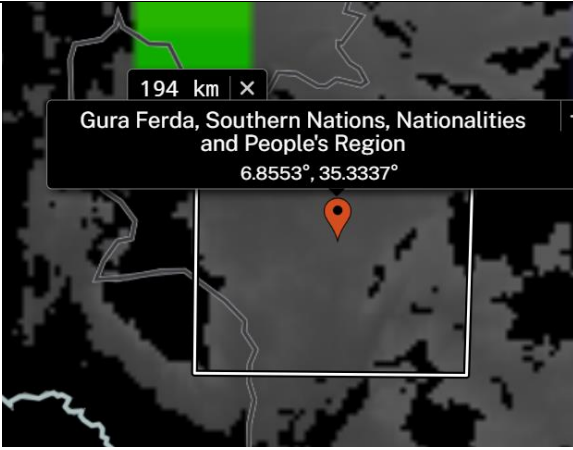
## Overview



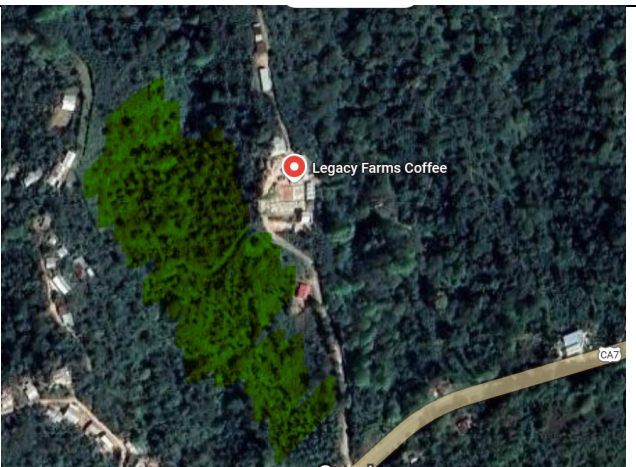
## Visual Confirmation



## Former SNNPR – Ethiopia [2,48%]

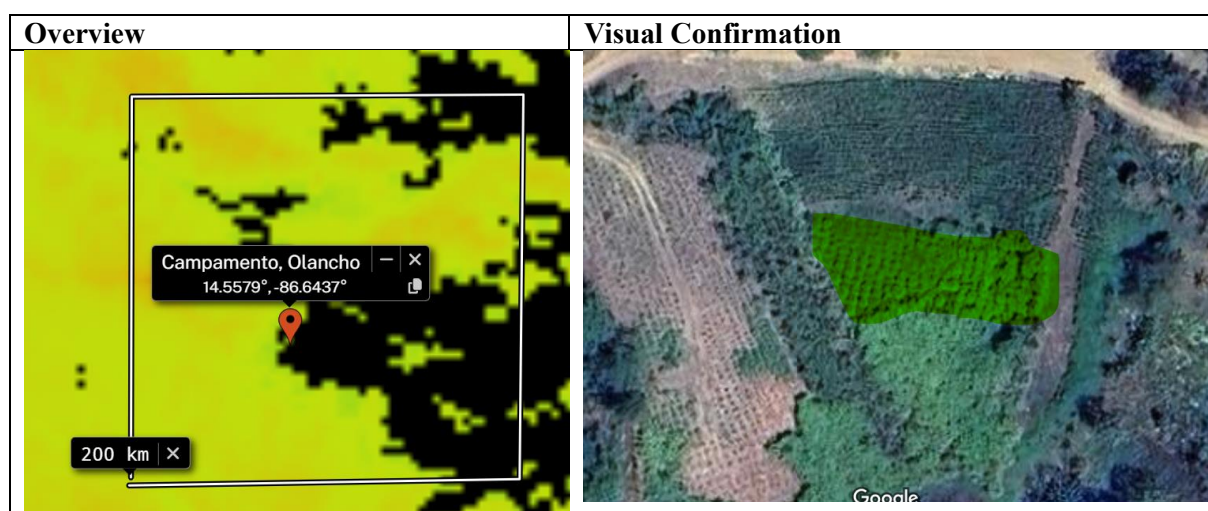
| Overview                                                                                                                                                                                              | Visual Confirmation                                                                                                                          |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Aleta Wondo North, Southern Nations, Nationalities and People's Region<br/>6.5759°, 38.4061°</p> <p>208 km X</p> |  <p>Mature AFS</p>                                         |
|  <p>Gura Ferda, Southern Nations, Nationalities and People's Region<br/>6.8553°, 35.3337°</p> <p>194 km X</p>       |  <p>Mature AFS shaded in Green (Mynsee Coffee Estate)</p> |

## Intibucá – Honduras [1,81%]

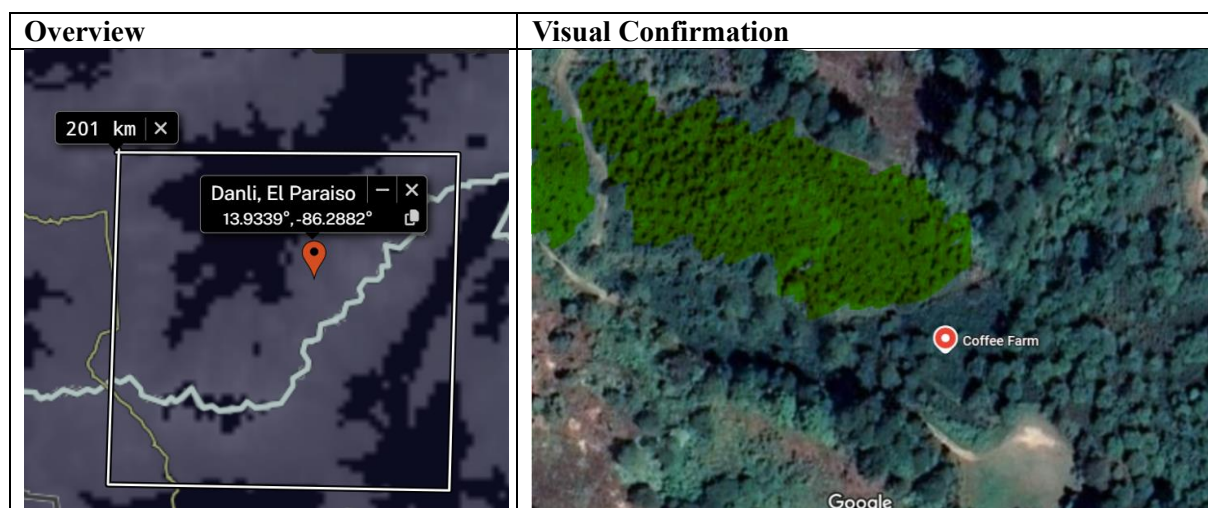
| Overview                                                                                                                                            | Visual Confirmation                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
|  <p>San Jose, La Paz<br/>14.2203°, -87.9285°</p> <p>201 km X</p> |  <p>Legacy Farms Coffee</p> |
| <p>Note: San Jose was used instead of Intibucá (Roughly 21 km away from each other)</p>                                                             |                                                                                                                 |



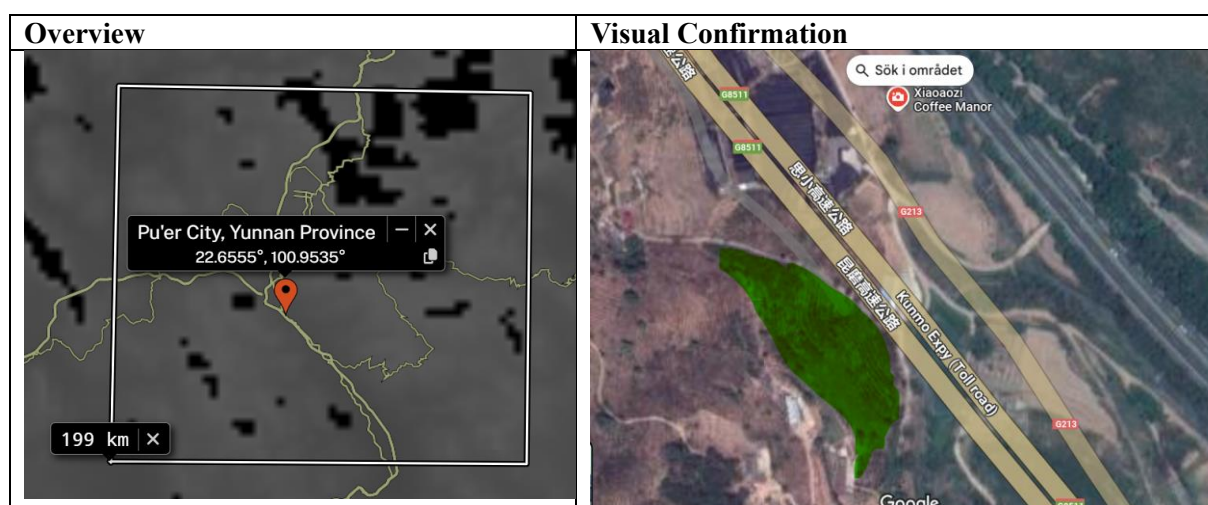
## Olancho – Honduras [1,81%]



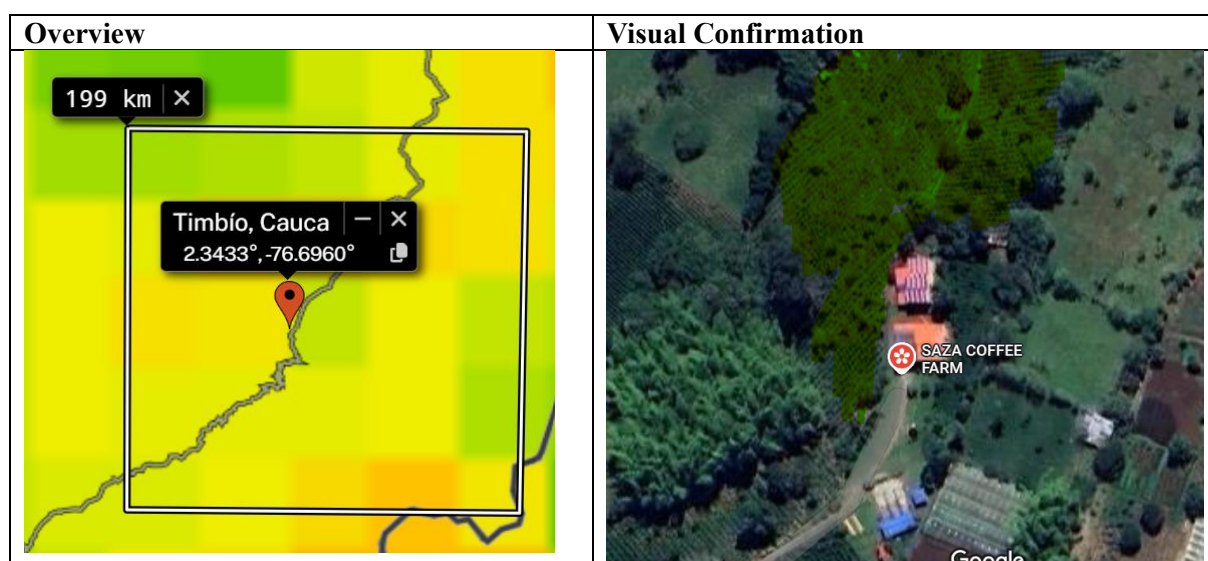
## El Paraíso Department – Honduras [1,81%]



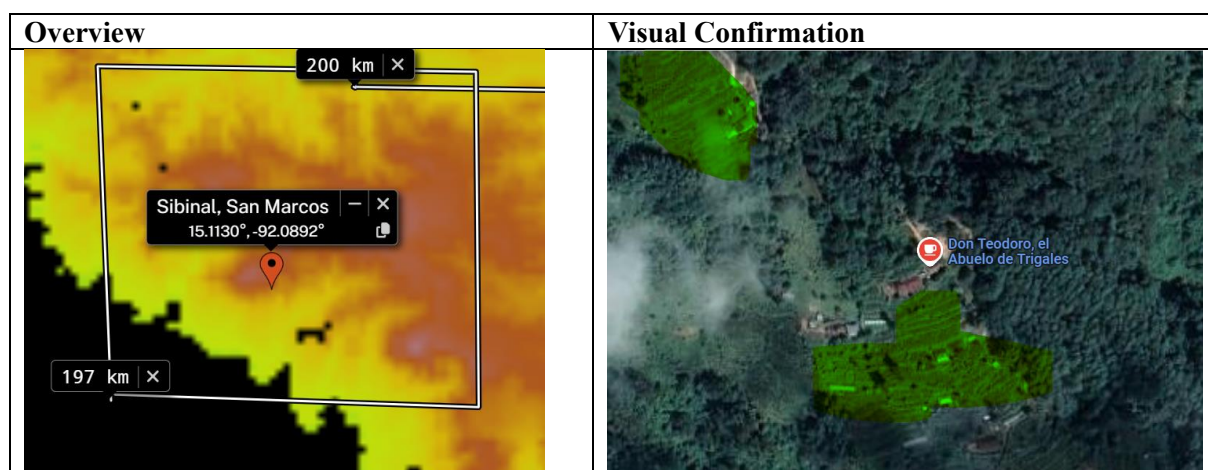
## Yunnan – China [1,80%]



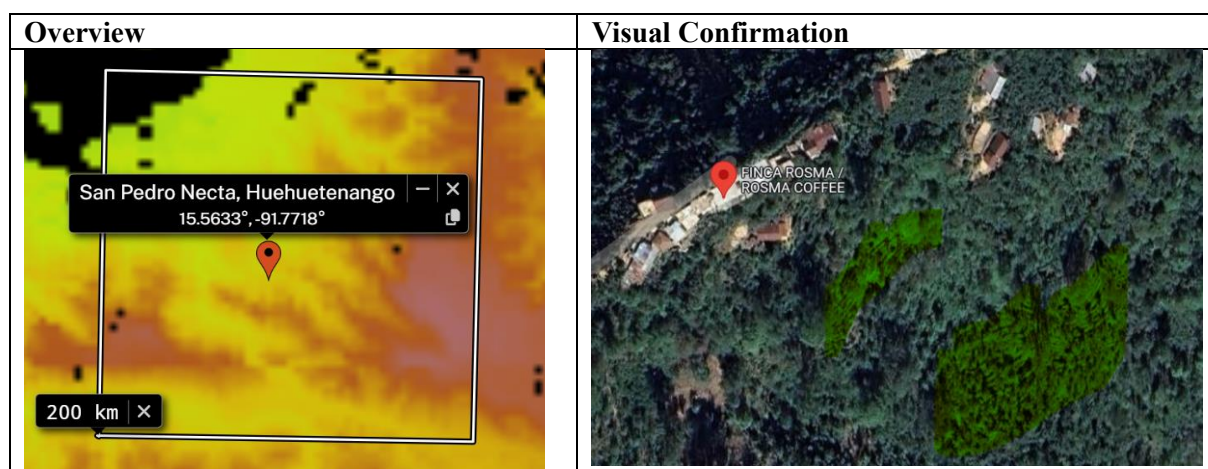
## Southern – Colombia [1,71%]



## San Marcos – Guatemala [1,39%]

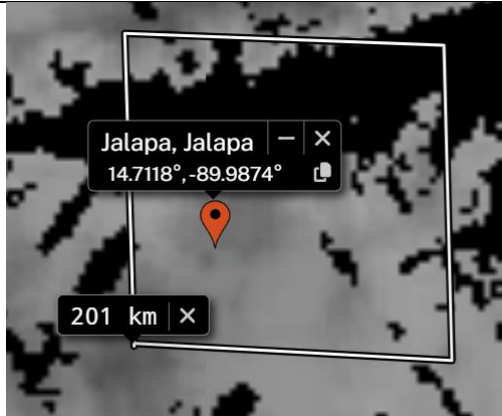


## Huehuetenango – Guatemala [1,29%]

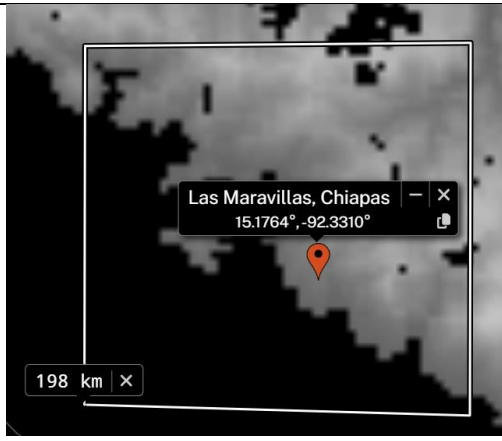




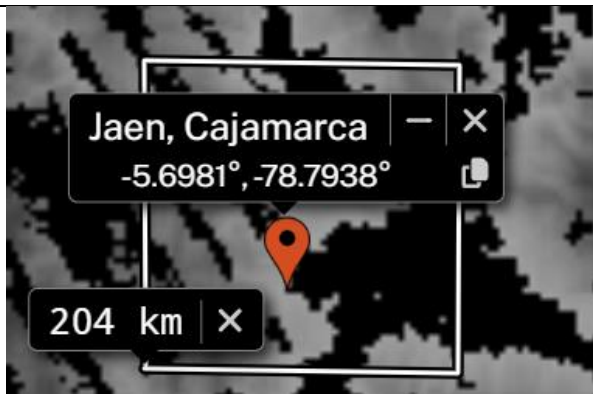
### Jalapa – Guatemala [1,11%]

| Overview                                                                                                                                                                                                                                                                                      | Visual Confirmation                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  An overview map showing the location of Jalapa, Guatemala. A red pin marks the location, with a text box displaying "Jalapa, Jalapa" and coordinates "14.7118°, -89.9874°". A scale bar indicates "201 km". |  <p>Mature AFS shaded in Green</p> A satellite image of the Jalapa area. The mature AFS (Agroforestry System) is shaded in green, showing a dense forested area. The text "Mature AFS shaded in Green" is visible below the image. |

### Chialapas – Mexico [1,29%]

| Overview                                                                                                                                                                                                                                                                                                      | Visual Confirmation                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  An overview map showing the location of Las Maravillas, Chiapas. A red pin marks the location, with a text box displaying "Las Maravillas, Chiapas" and coordinates "15.1764°, -92.3310°". A scale bar indicates "198 km". |  A satellite image of the Las Maravillas area. The image shows a dense forested area with a winding road or path. The text "Google" is visible at the bottom of the image. |

### Cajamarca – Peru [0,98%]

| Overview                                                                                                                                                                                                                                                                                       | Visual Confirmation                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  An overview map showing the location of Jaen, Cajamarca. A red pin marks the location, with a text box displaying "Jaen, Cajamarca" and coordinates "-5.6981°, -78.7938°". A scale bar indicates "204 km". |  <p>Ridges in green</p> A satellite image of the Jaen area. The image shows a dense forested area with a winding road or path. The text "Ridges in green" is visible below the image. Other labels in the image include "Rio Arroyo", "PARQ FAMIL", and "Iglesia San". |



## Exchange Rate Information

| Country          | Currency Name           | Currency Code | Included (As of version 1.0) |
|------------------|-------------------------|---------------|------------------------------|
| <b>Brazil</b>    | Brazilian Real          | BRL           | Yes <sup>36</sup>            |
| <b>Colombia</b>  | Colombian Peso          | COP           | No                           |
| <b>Ethiopia</b>  | Ethiopian Birr          | ETB           | No                           |
| <b>Honduras</b>  | Honduran Lempira        | HNL           | No                           |
| <b>Guatemala</b> | Guatemalan Quetzal      | GTQ           | No                           |
| <b>Peru</b>      | Peruvian Sol            | PEN           | No                           |
| <b>China</b>     | Chinese Yuan (Renminbi) | CNY           | Yes <sup>37</sup>            |
| <b>Mexico</b>    | Mexican Peso            | MXN           | Yes <sup>38</sup>            |

The only currency implemented as of version 1.0 are those that could be easily extracted via the use of APIs. It might be possible to extract currency from the respective countries' central bank, but it will take without a doubt quite a while. Exchange rates information was extracted using Saint Louis API.

## Futures as Proxies for Arabica Spot Prices

### Missing Values

In the original data set, over 2780 observations were missing, corresponding to 1273 Saturday +1273 Sunday), holidays (234), and other events (16). While this was fine when the main focus was on the effect of future prices on exchange rates and spot prices have on each other, skipping days when the weather is the major variable of interest is not wise, forcing interpolation on the missing values when it comes to financial data.

### How similar are futures to spot prices if used as a proxy? (Based on Earlier Data Set)

| TERM                | ESTIMATE | STD. ERROR | T VALUE | P-VALUE    | INTERPRETATION                                                                                                                                             |
|---------------------|----------|------------|---------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (INTERCEPT)         | -16.54   | 0.296      | -55.86  | <2e-16 *** | Spot price tends to be about 16.5 units less than futures price baseline, all else zero. Reflects contango bias or systematic difference.                  |
| PRICE_FUTURES_LAG1  | 0.694    | 0.011      | 63.38   | <2e-16 *** | For every 1 unit increase in yesterday's futures price, spot price today increases by 0.694 units — a strong and highly significant positive relationship. |
| PRICE_FUTURES_LAG7  | 0.032    | 0.015      | 2.15    | 0.0318 *   | A smaller but statistically significant positive effect from the futures price a week ago (7 days). Suggests some lingering influence.                     |
| PRICE_FUTURES_LAG14 | 0.016    | 0.014      | 1.12    | 0.263      | Not statistically significant — futures price from 2 weeks ago has little independent effect on spot price after controlling for other lags.               |
| PRICE_FUTURES_LAG21 | 0.022    | 0.014      | 1.55    | 0.122      | Also not significant — futures price from 3 weeks ago is weak in explaining spot price here.                                                               |
| PRICE_FUTURES_LAG28 | 0.139    | 0.010      | 13.55   | <2e-16 *** | A substantial and highly significant effect from futures price 4 weeks ago. This suggests some delayed adjustment or longer-term market memory.            |

<sup>36</sup> Board of Governors of the Federal Reserve System (US), Brazilian Reals to U.S. Dollar Spot Exchange Rate [DEXBZUS], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/DEXBZUS>, July 21, 2025.

<sup>37</sup> Board of Governors of the Federal Reserve System (US), Chinese Yuan Renminbi to U.S. Dollar Spot Exchange Rate [DEXCHUS], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/DEXCHUS>, July 21, 2025

<sup>38</sup> Board of Governors of the Federal Reserve System (US), Mexican Pesos to U.S. Dollar Spot Exchange Rate [DEXMXUS], retrieved from FRED, Federal Reserve Bank of St. Louis; <https://fred.stlouisfed.org/series/DEXMXUS>, July 21, 2025.

# Conclusions and Streamlining Commodity Research

AI tools tend to hallucinate and often refer to things that do not exist, or non-official sources. It might be possible to use the Ellmer package in R and LLM in order to analyze and extract the most important information based on a limited set of documents, potentially limiting hallucinations, and also grading those documents that are most worth reading. Here are the steps that might streamline the process.

- 1) Gathering Reports from the USDA and other reputable sources
- 2) Use Ellmer to extract and grade the report most worthy of reading
- 3) Determine the areas with a greater impact of global production
- 4) Reutilize the code from the data set to extract information from Nasa Power



